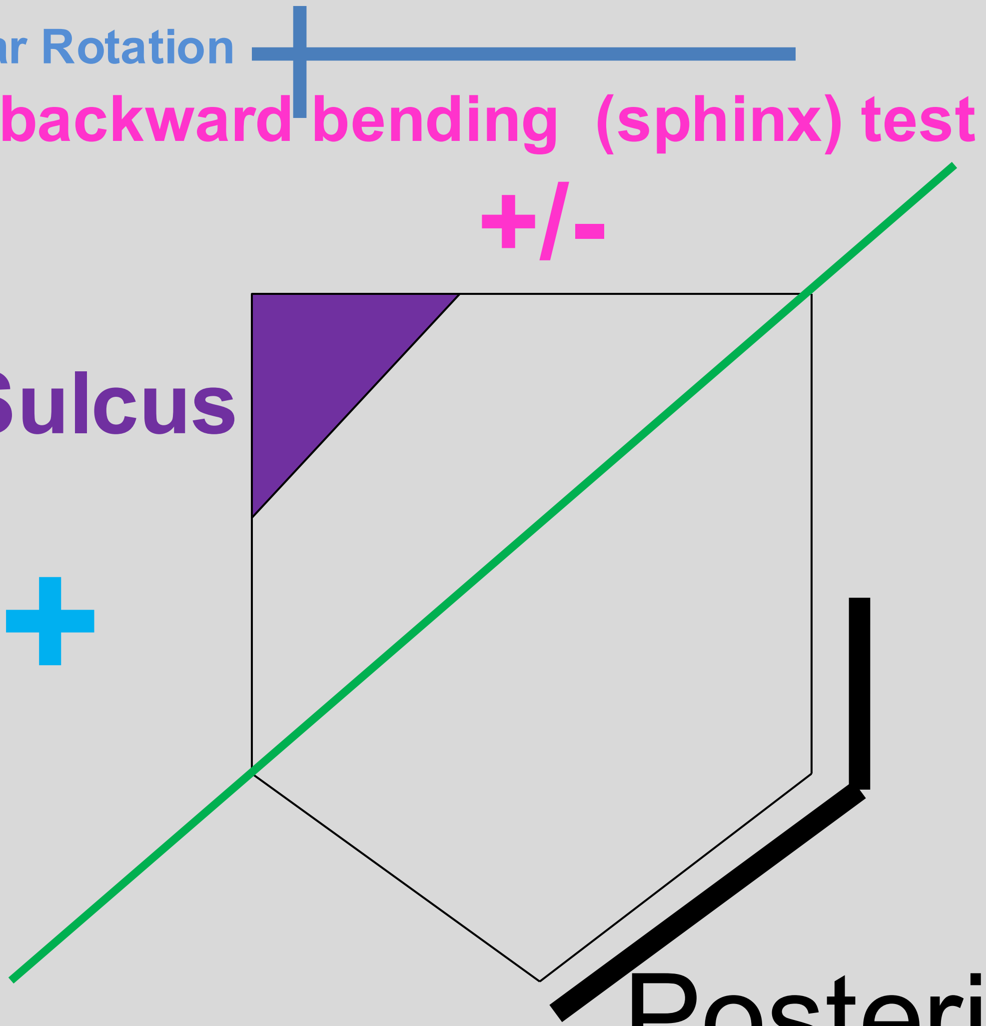
Diagramming Sacral Diagnosis



- Using the notation from the previous slide, diagram the following sacra according to the physical exam findings in the left lower corners.
- Then try to figure out the diagnosis
- All this will be demonstrated in lecture so if you are having trouble, don't worry.

- Also note, the following slides do not mention an L5 diagnosis.
- If L5 is rotated the same direction as the sacrum, then the diagnosis is a “sacral rotation”
  - Example: right on right forward sacral **rotation**
- If L5 is rotated the opposite direction as the sacrum **OR** if there is no L5 dysfunction, then the diagnosis is a “sacral torsion”
  - Example: right on right forward sacral **torsion**

Work Slides

- Just to be clear:

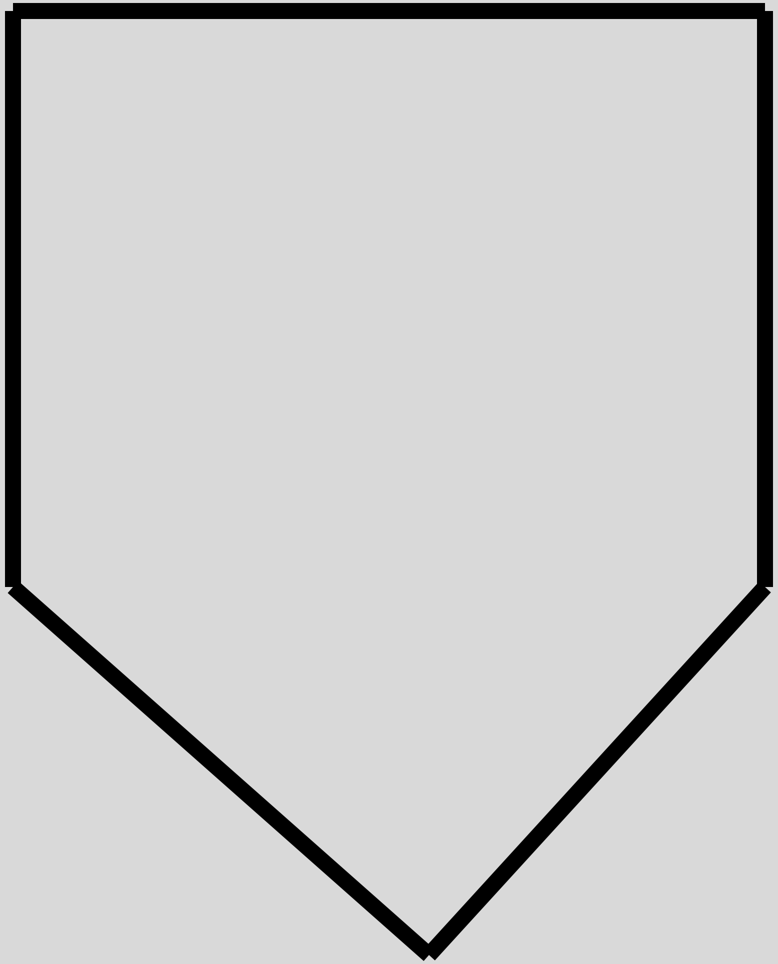
  Work Slides 1 – 10 do not mention an L5 diagnosis. You will need L5 to determine a torsion vs. a rotation. The following examples are here for you to work out the sacral part of the diagnosis.

  When studying, add in the L5 diagnosis to make it more interesting.

- **Work slides 11 & 12 do mention an L5 diagnosis so you can practice making a sacral diagnosis given ONLY the L5 diagnosis.**
- Also, the abbreviation “bbt”, which is NOT an officially accepted abbreviation, is used for “backward bending test” on the following slides.

Work slide #1

L5
diagnosis?



-Negative Spring test

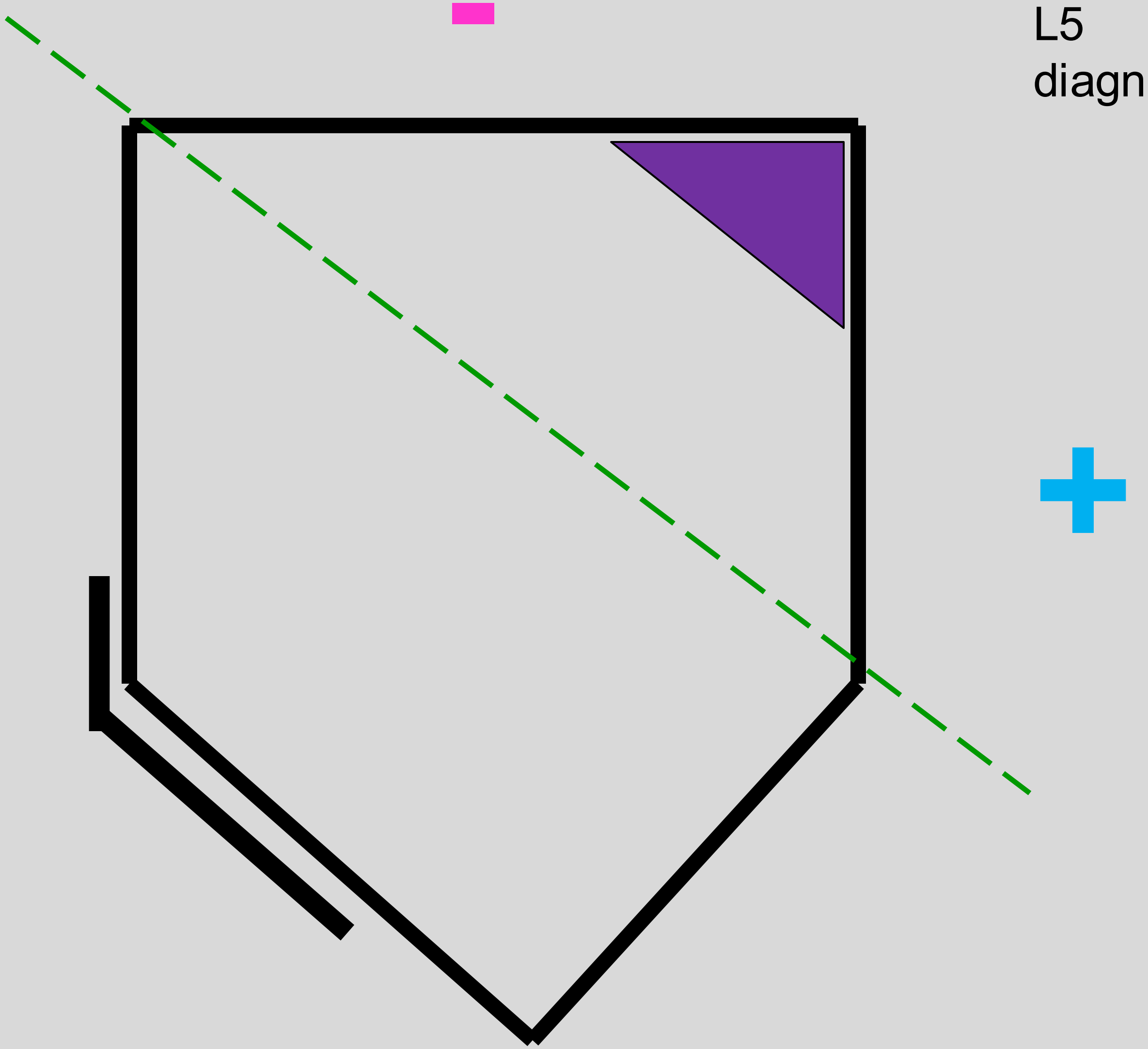
-Neg. bbt /sphinx test

-Deep sulcus on right

-Posterior/Inferior ILA on the left

-Seated Flexion???

Work slide #1: **Left on Left Forward sacrum**

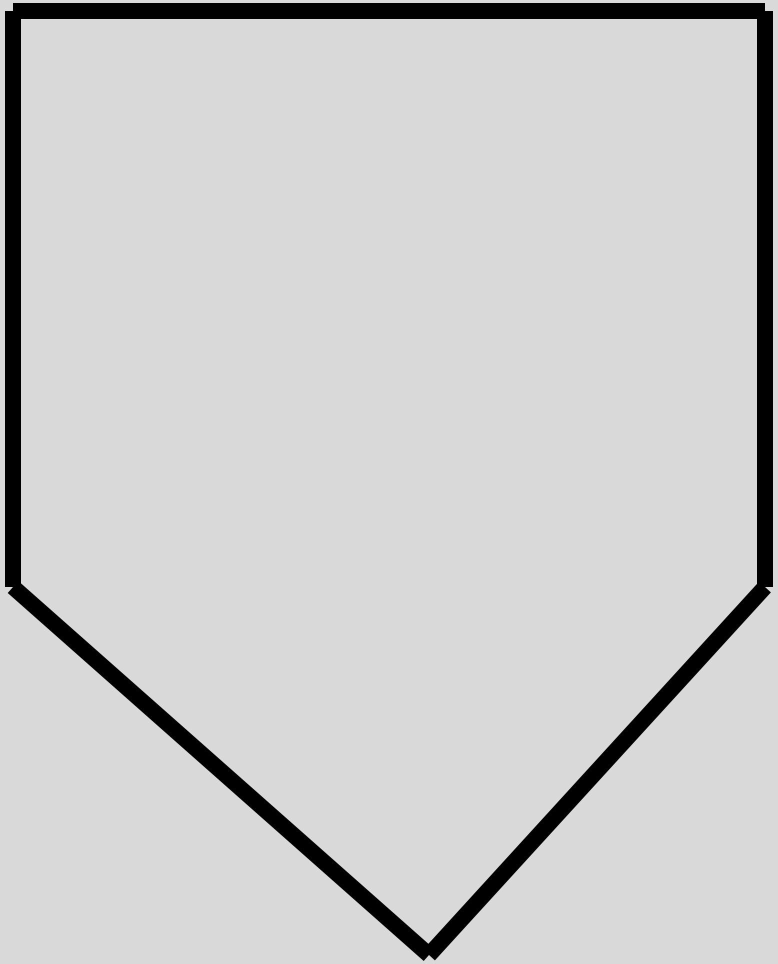


L5
diagnosis?

- -Negative Spring test
- -Neg. bbt /sphinx test
- -Deep sulcus on right
- -Posterior/Inferior ILA on the left
- -Seated Flexion???

Work slide #2

L5 diagnosis?



-Positive Spring test

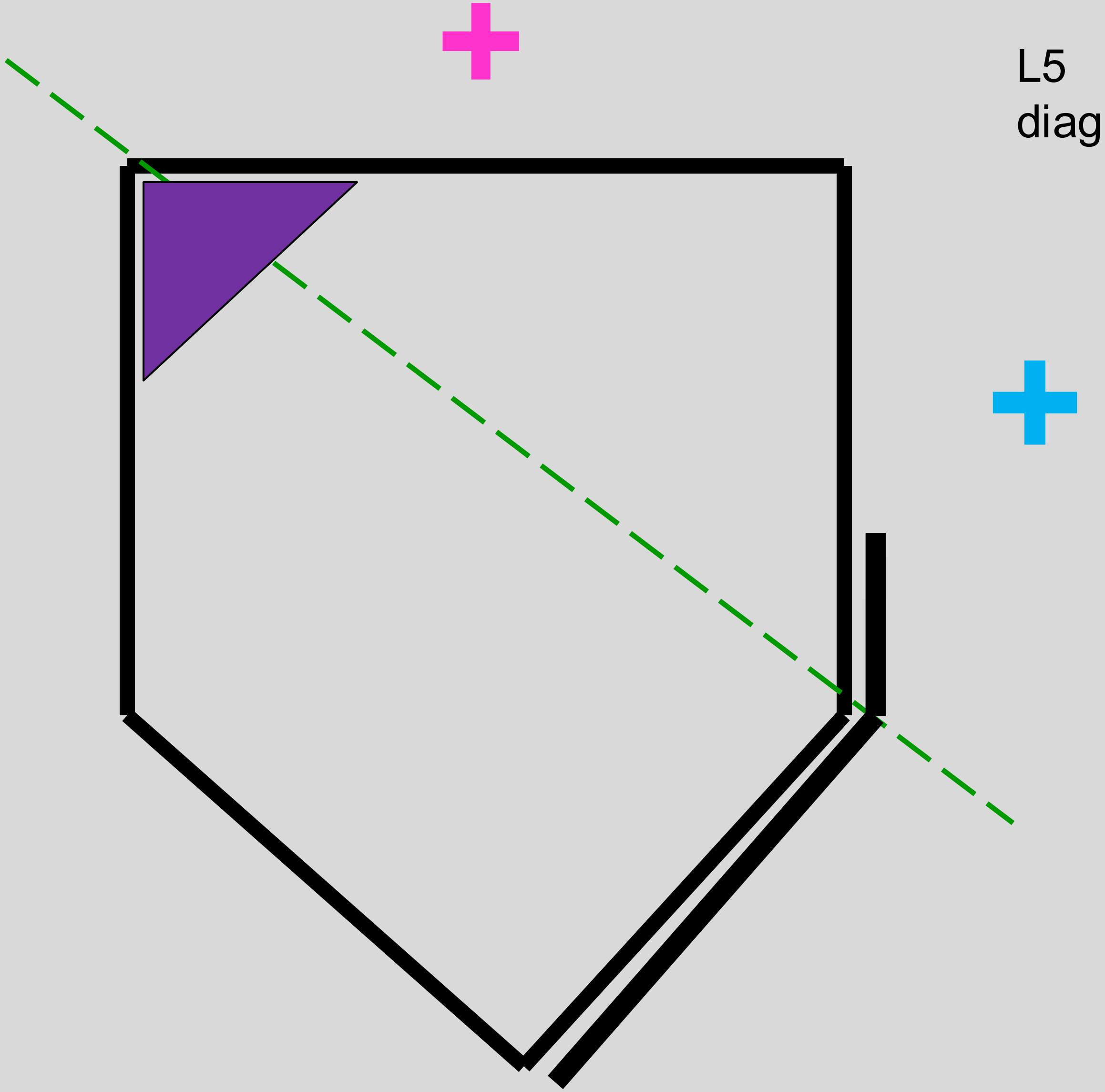
-Pos. bbt / sphinx test

-Deep sulcus on left

-Posterior/Inferior ILA on the right

-Seated flexion test??

Work Slide #2: **Right on Left backward sacrum**



L5 diagnosis?

-Positive Spring test

-Pos. bbt / sphinx test

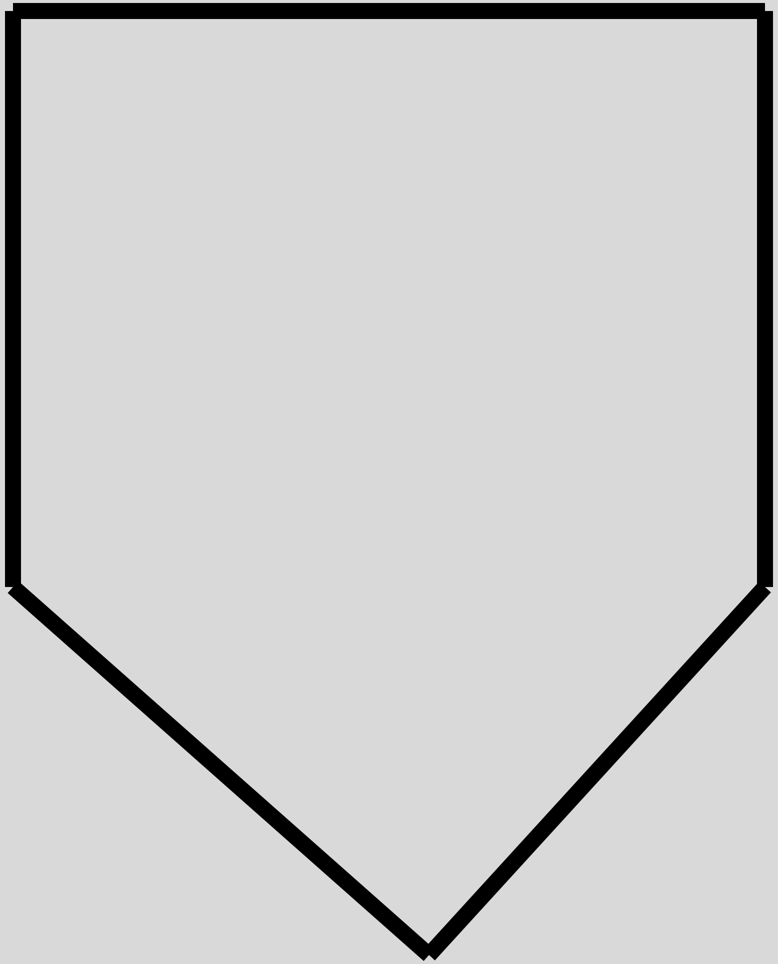
-Deep sulcus on left

-Posterior/Inferior ILA on the right

-Seated flexion test??

Work slide #3

L5
diagnosis?



-Negative Spring test

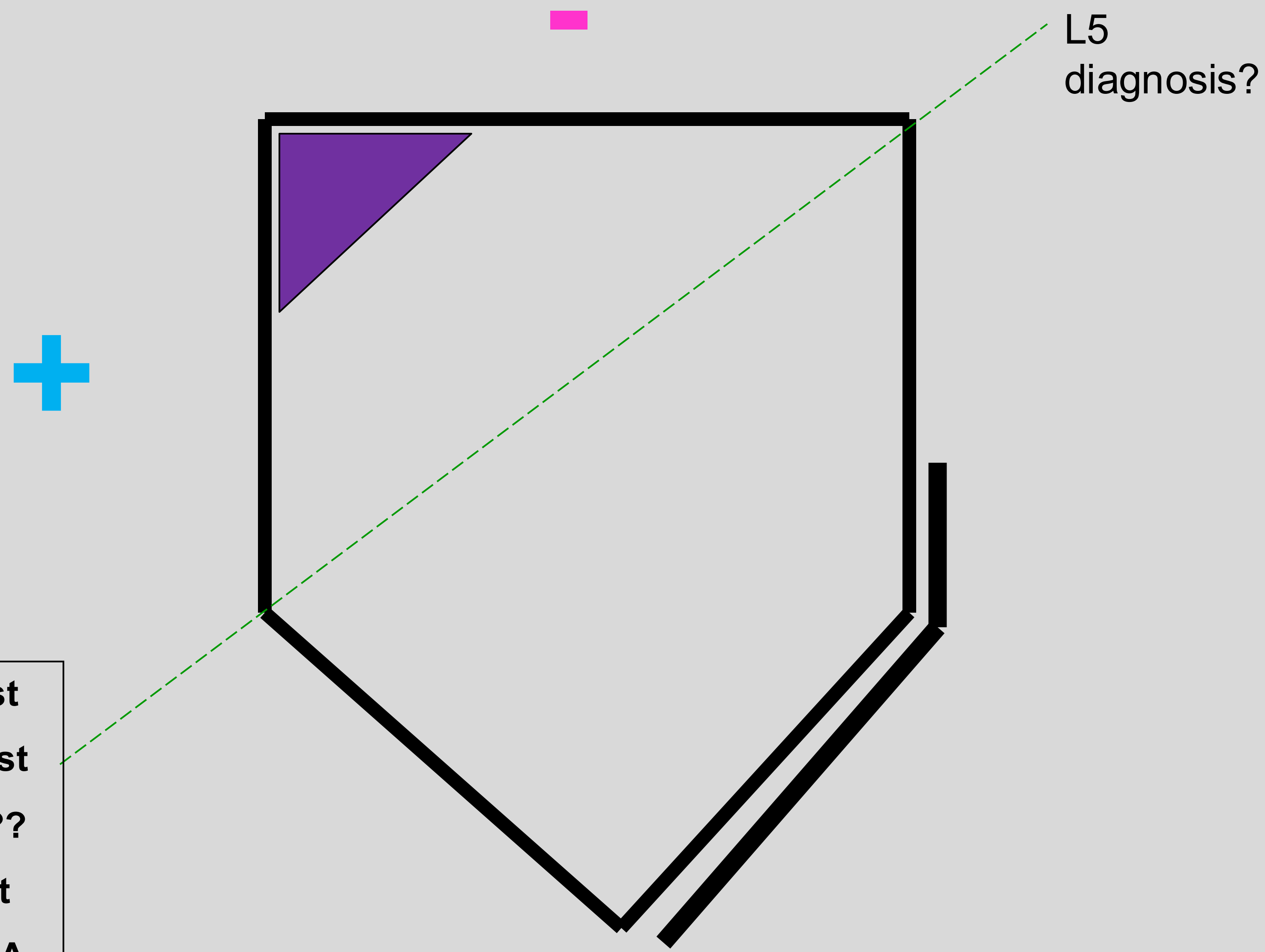
-Neg. bbt / sphinx test

-Seated flexion test??

-Deep sulcus on left

-Posterior/Inferior ILA on the right

Work Slide #3: **Right on Right Forward Sacrum**



-Negative Spring test

-Neg. bbt / sphinx test

-Seated flexion test??

-Deep sulcus on left

-Posterior/Inferior ILA on the right

Work slide #4

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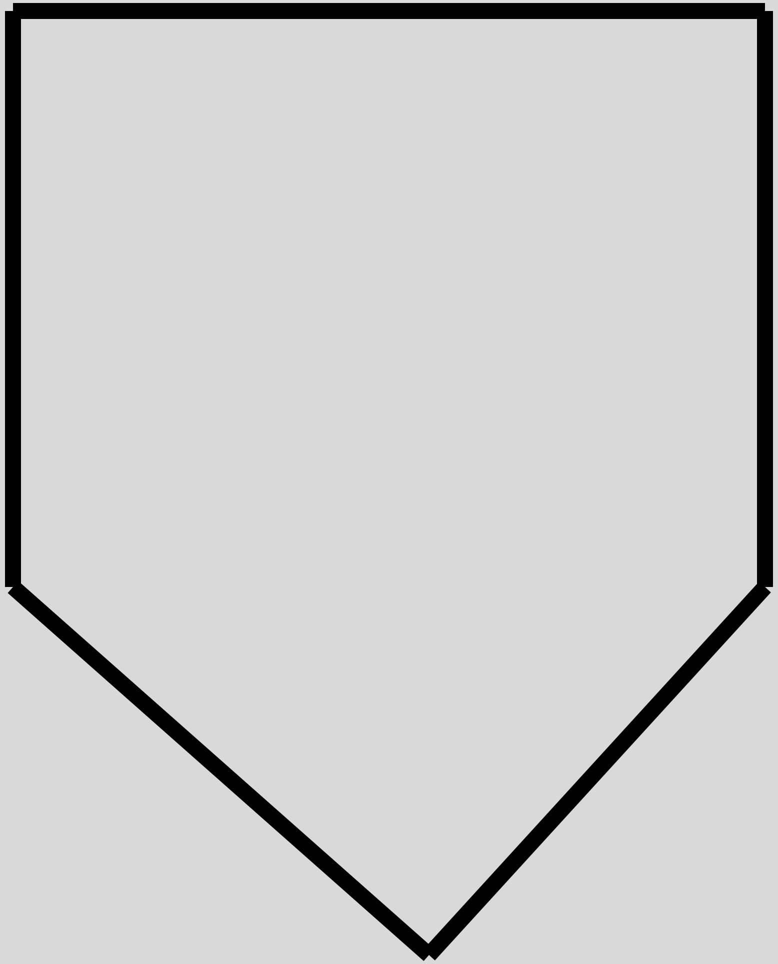
L5
diagnosis?



-Positive Spring test

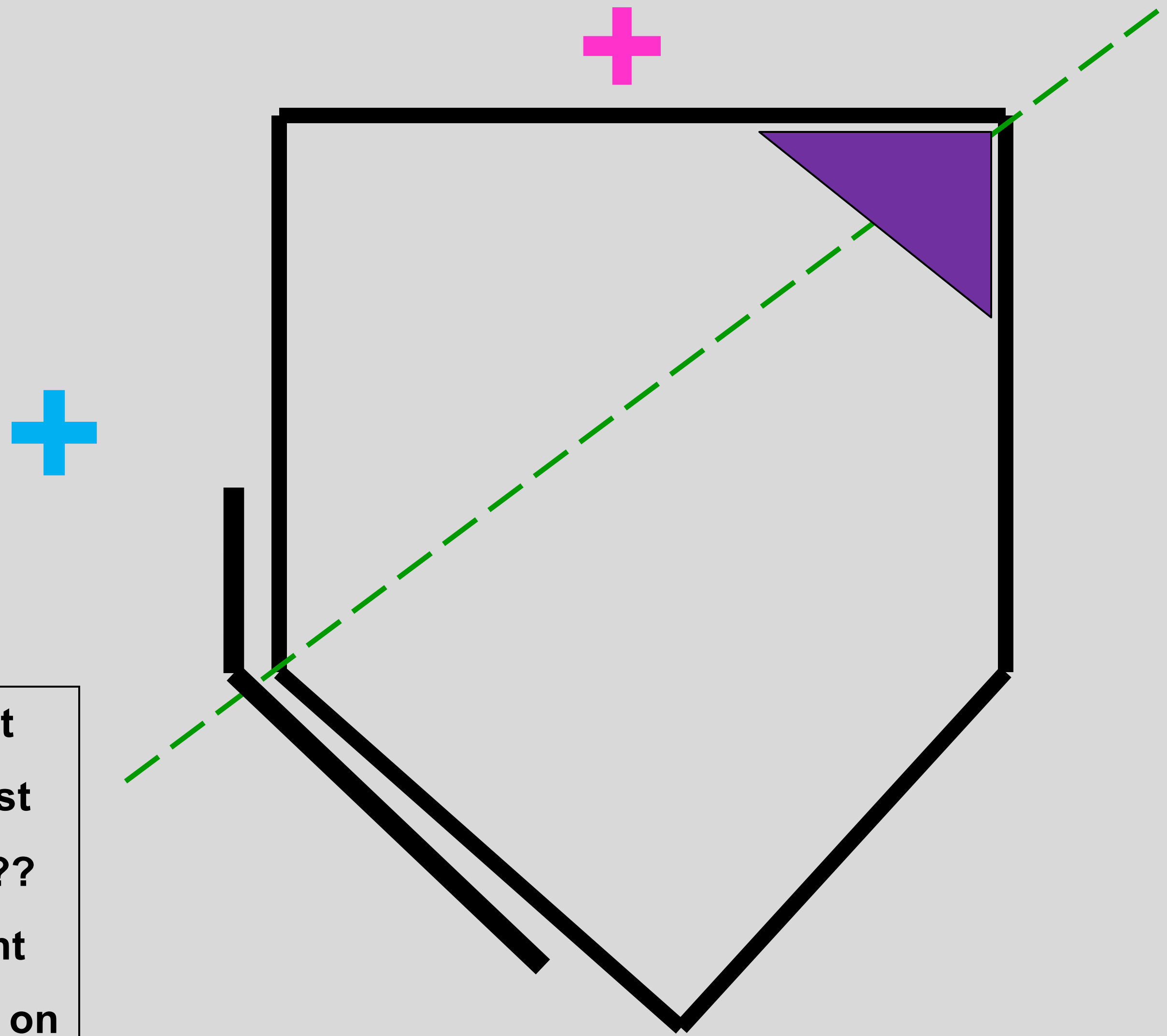
-Pos. bbt / sphinx test

-Seated Flexion test??

-Deep sulcus on right

-Posterior/Inferior ILA on the left

Work Slide #4: Left on Right Backward Sacrum



-Positive Spring test

-Pos. bbt / sphinx test

-Seated Flexion test??

-Deep sulcus on right

-Posterior/Inferior ILA on the left

Work slide #5

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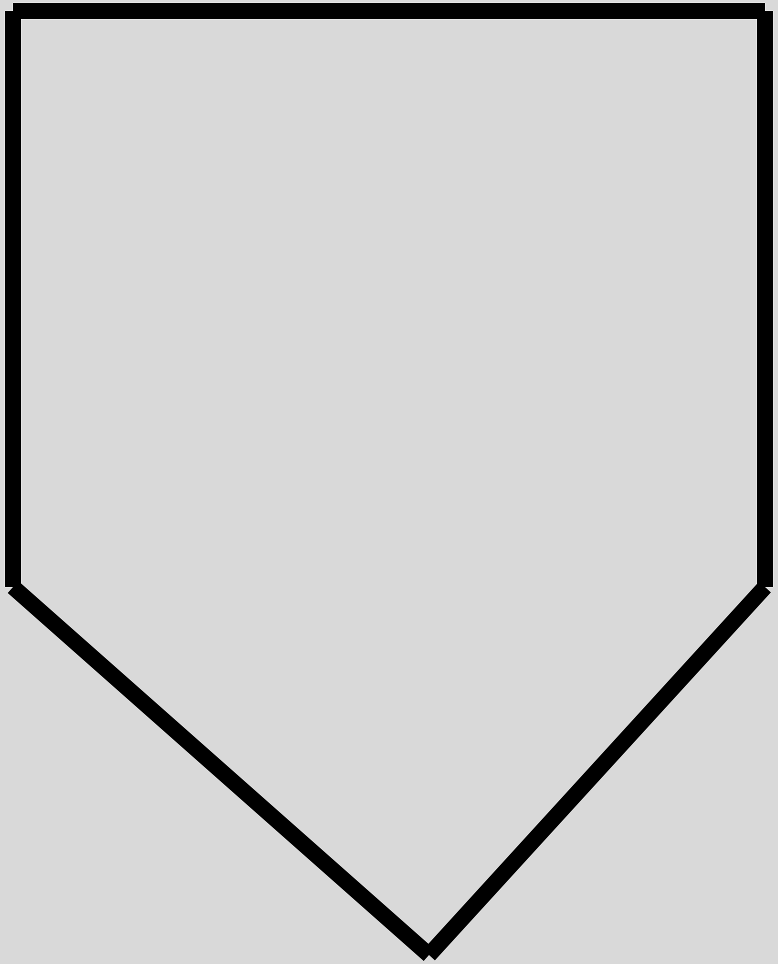
L5
diagnosis?



- positive seated flexion test on left
- Neg. spring test
- Neg. bbt /sphinx test
- Deep sulcus on left
- Post. /inferior ILA on left

Work Slide # 5: Left Unilateral Sacral Flexion

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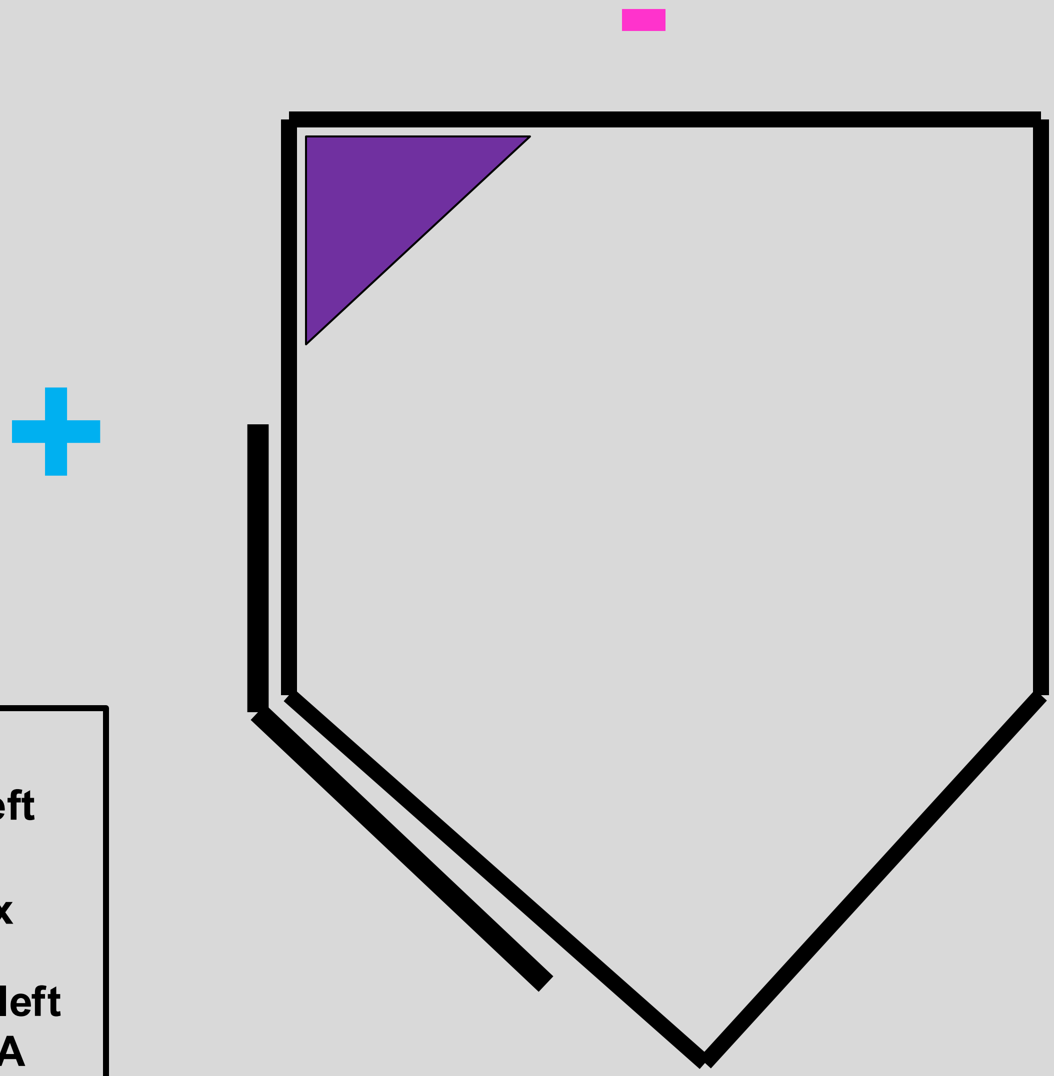
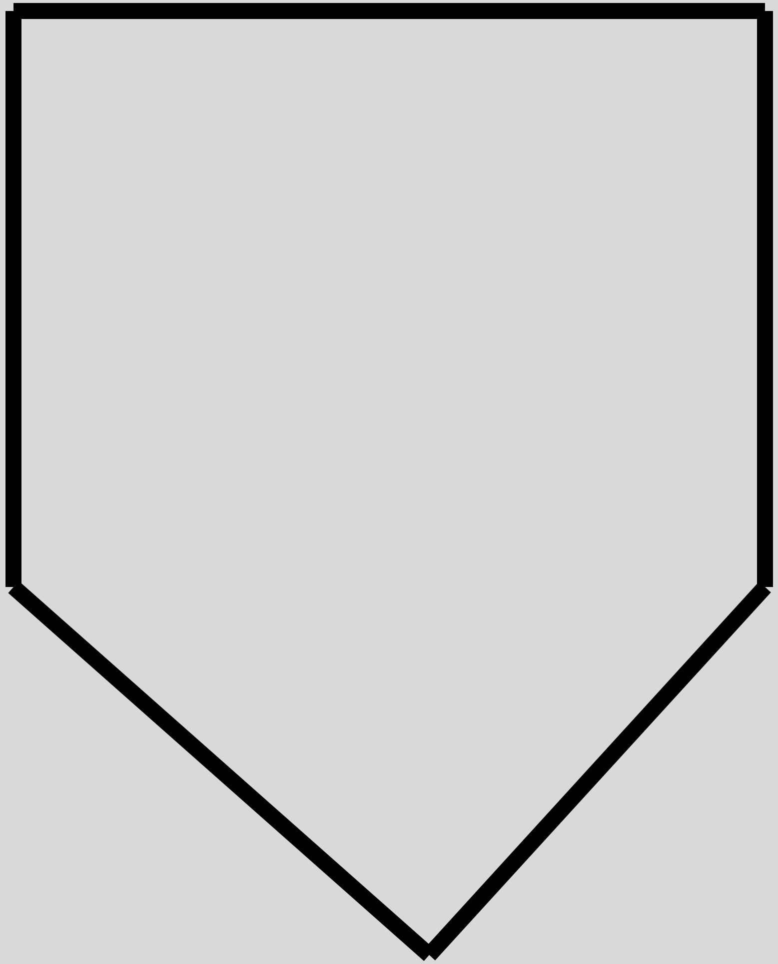
L5 diagnosis?



- positive seated flexion test on left
- Neg. spring test
- Neg. bbt /sphinx test
- Deep sulcus on left
- Post. /inferior ILA on left

Work slide #6

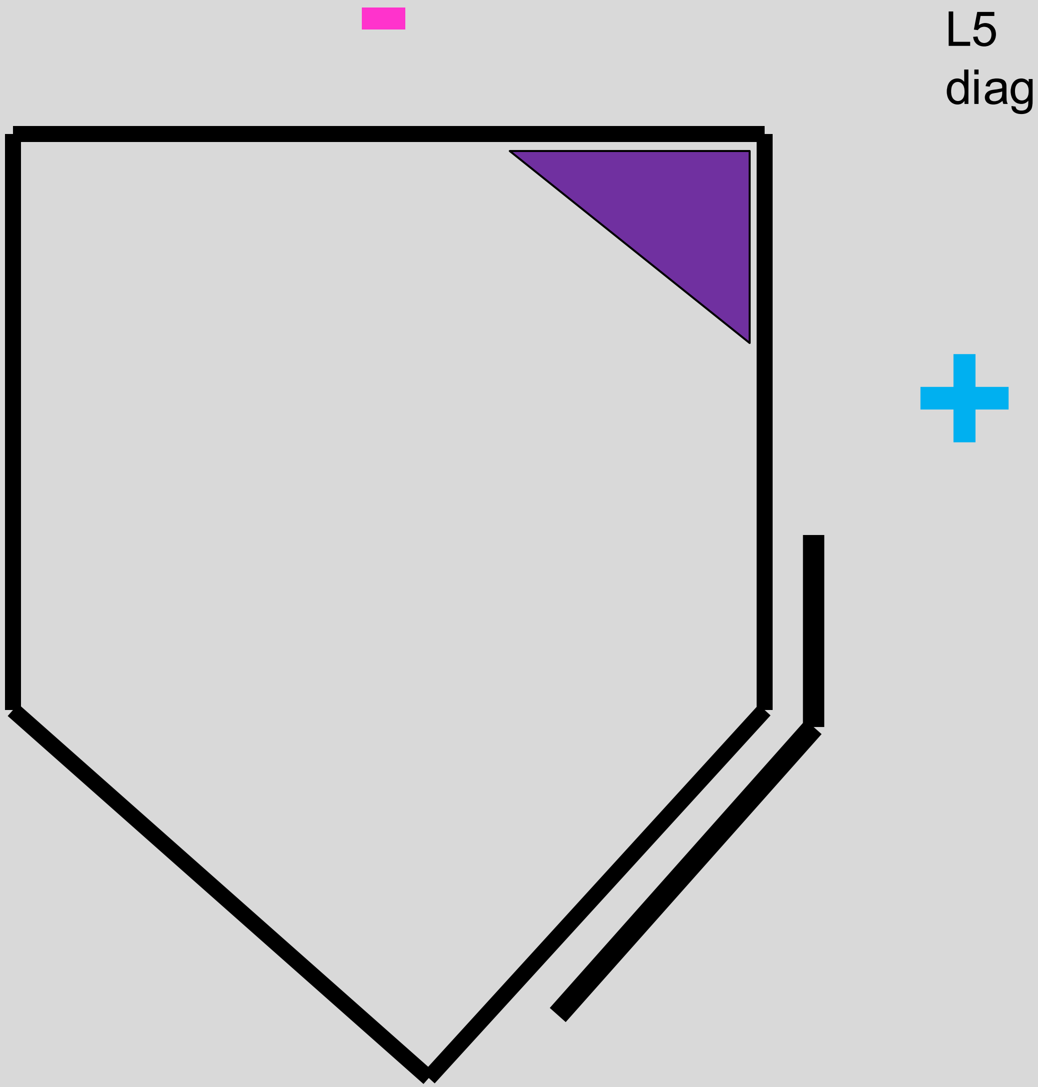
L5
diagnosis?



- positive seated flexion test on right
- Neg. spring test
- Neg. bbt / sphinx test
- Deep sulcus on right
- Post. /inferior ILA on right

Work Slide # 6: **Right unilateral Sacral Flexion**

L5 diagnosis?



- positive seated flexion test on right
- Neg. spring test
- Neg. bbt / sphinx test
- Deep sulcus on right
- Post. /inferior ILA on right

Work slide #7

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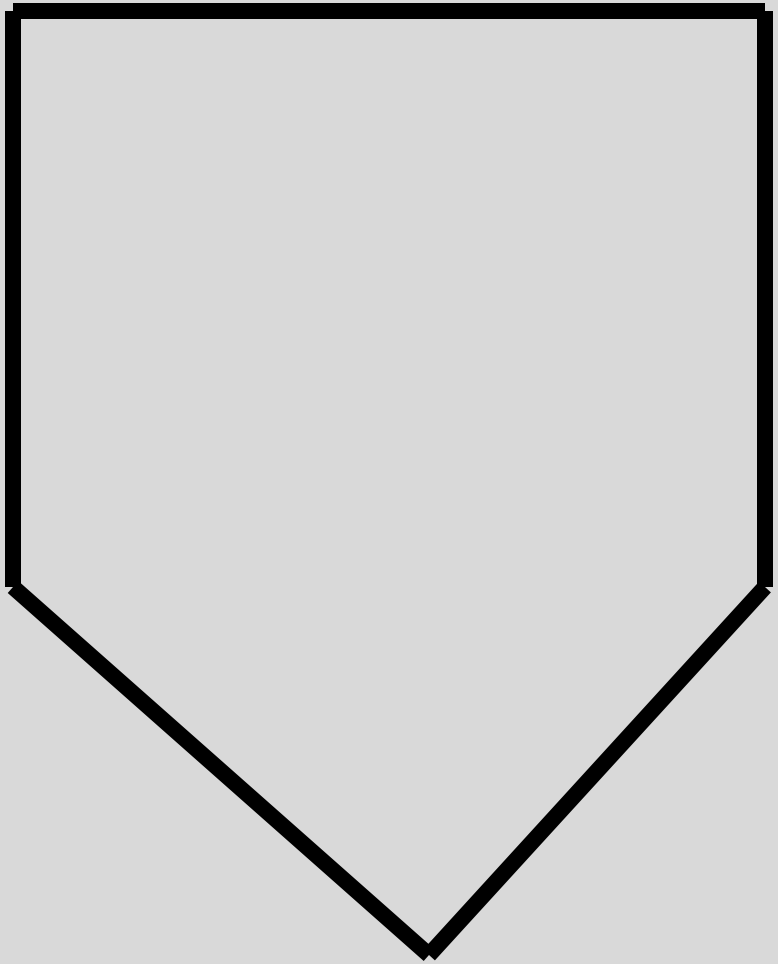
L5
diagnosis?



- shallow sulcus on the right
- anterior/superior ILA on right
- positive seated flexion test on right
- Pos. bbt / sphinx test
- Pos. spring test

Work Slide # 7: Right Unilateral Sacral Extension

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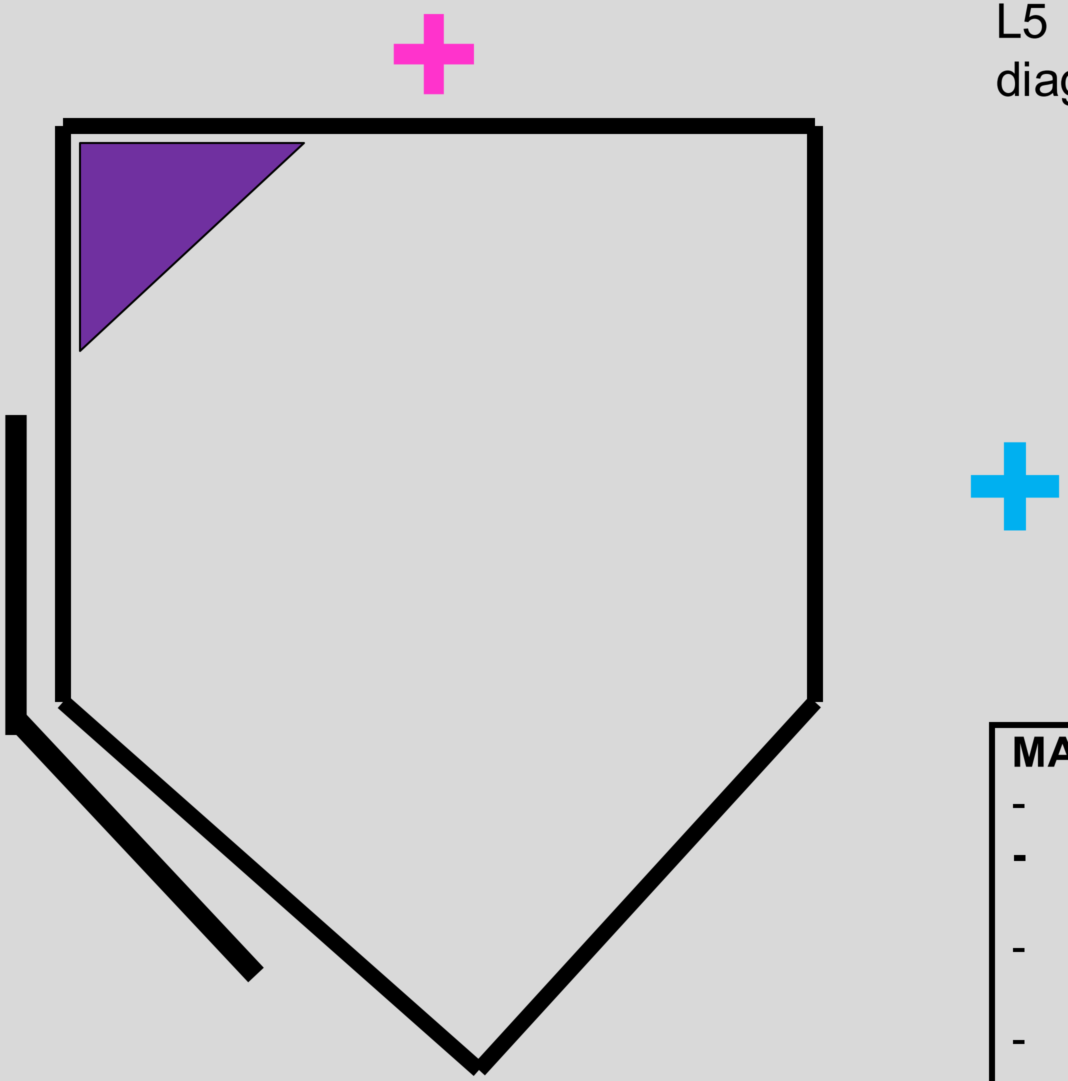


L5 diagnosis?

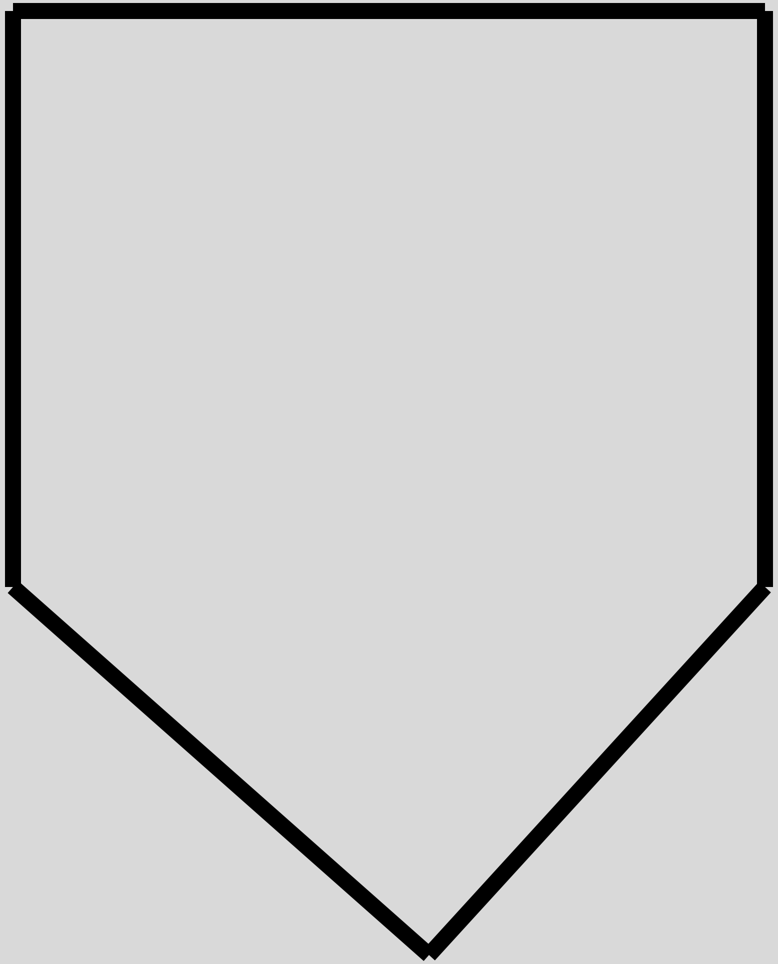
- shallow sulcus on the right
- anterior/superior ILA on right
- positive seated flexion test on right
- Pos. bbt / sphinx test
- Pos. spring test

MAY ALSO BE GIVEN AS:

- Deep sulcus on the left
- Posterior/Inferior ILA on left
- positive seated flexion test on right
- Pos. sphinx test
- Pos. spring test

Work slide #8

L5
diagnosis?



- shallow sulcus on the left
- anterior/superior ILA on left
- positive seated flexion test on left
- Pos. bbt / sphinx test
- Pos. spring test

Work Slide # 8: Left Unilateral Sacral Extension

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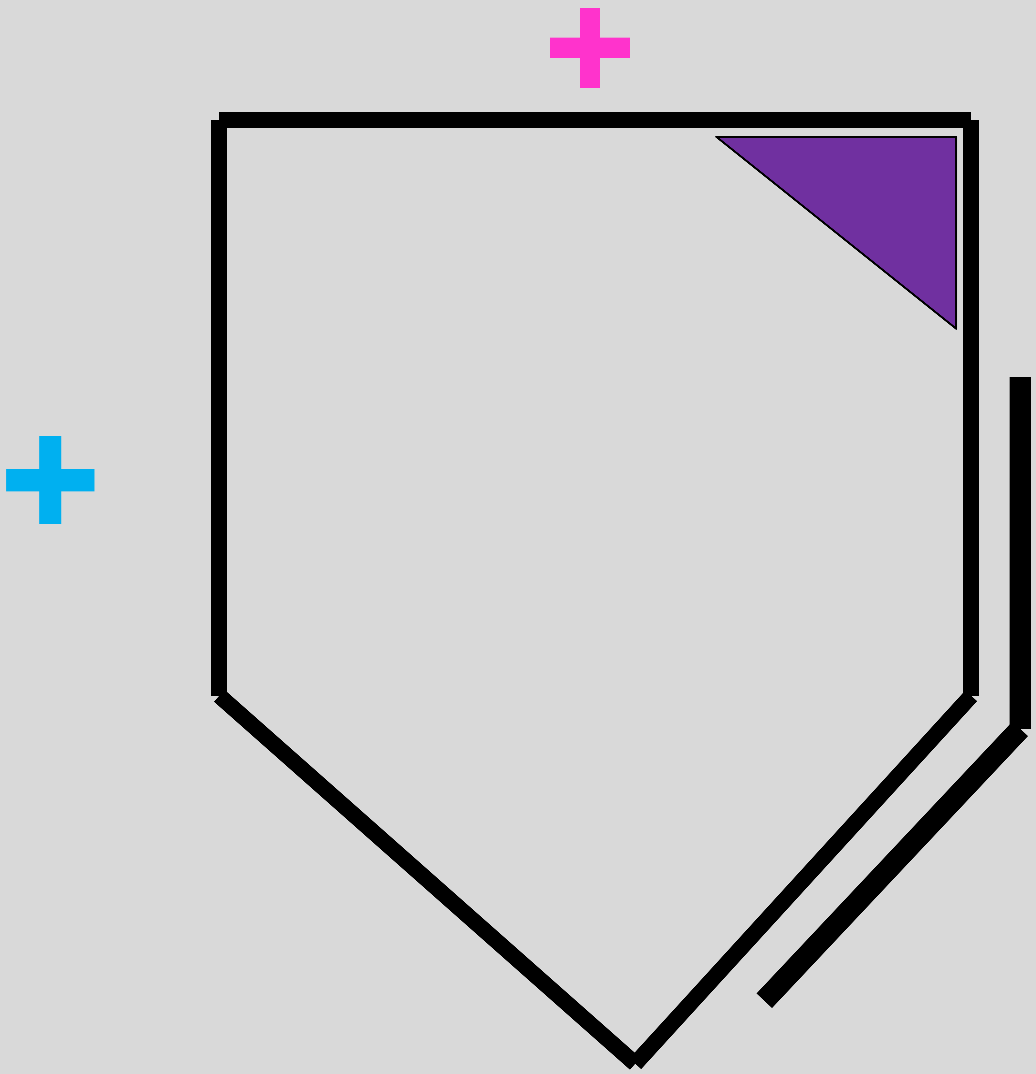
L5 diagnosis?

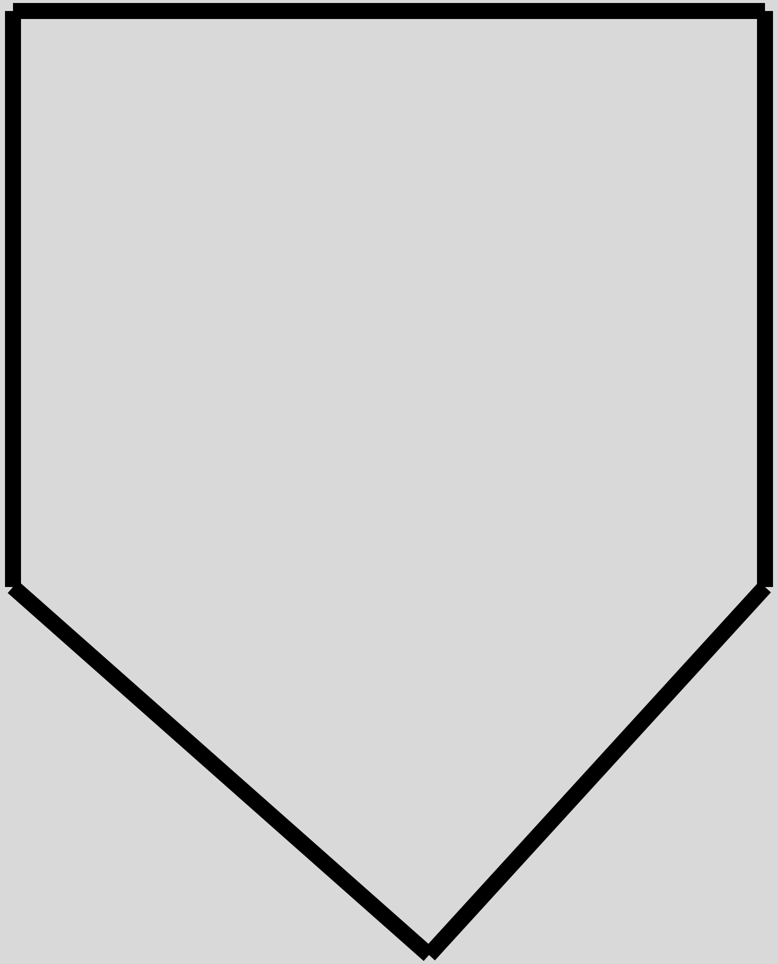


- shallow sulcus on the left
- anterior/superior ILA on left
- positive seated flexion test on left
- Pos. bbt / sphinx test
- Pos. spring test

MAY ALSO BE GIVEN AS:
- Deep sulcus on the right
- Posterior/Inferior ILA on right
- positive seated flexion test on left
- Pos. sphinx test
- Pos. spring test

Work slide #9

L5 diagnosis?



- Negative spring test
- B/L deep sulci
- B/L posterior ILAs

Work Slide # 9: Bilateral Sacral Flexion

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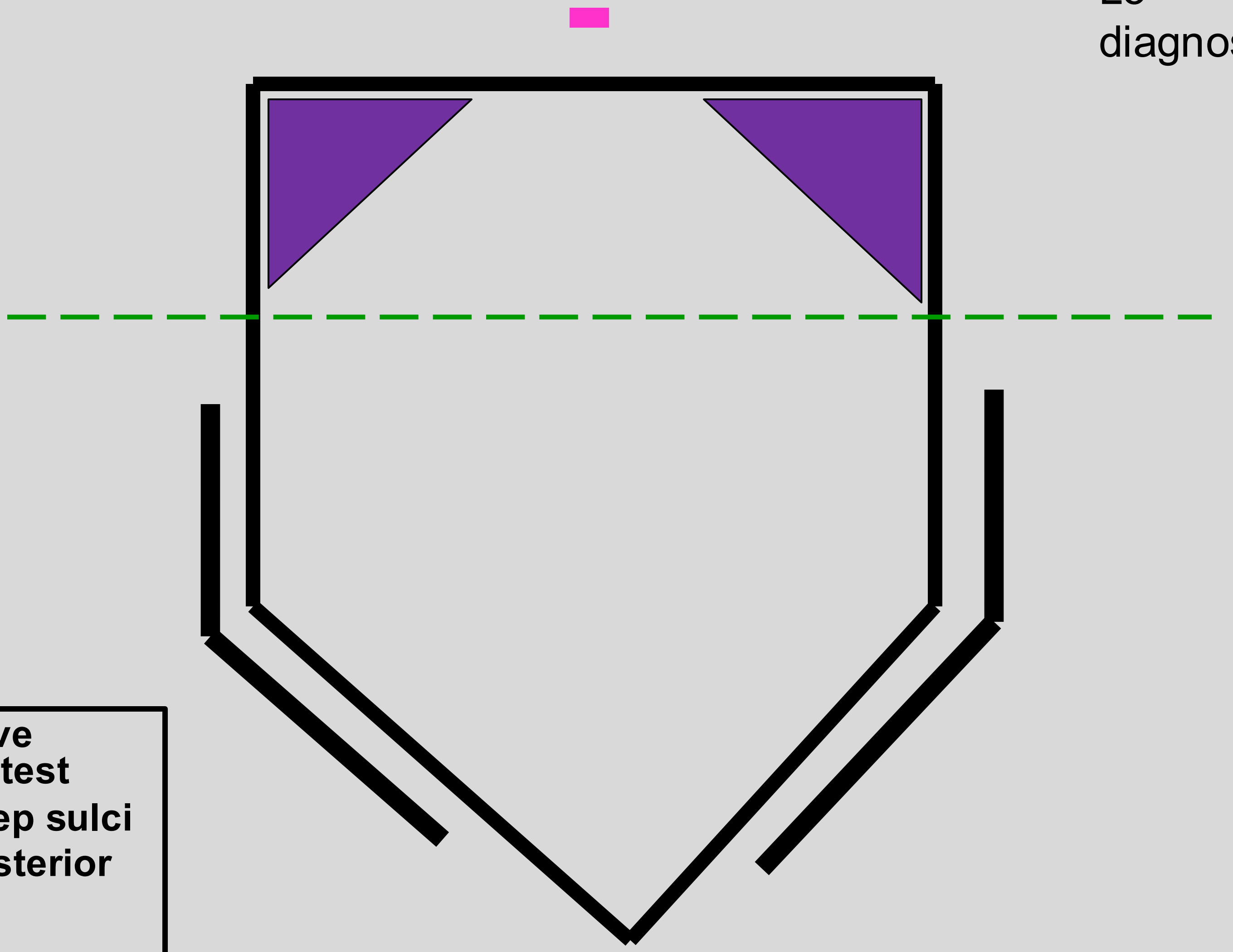
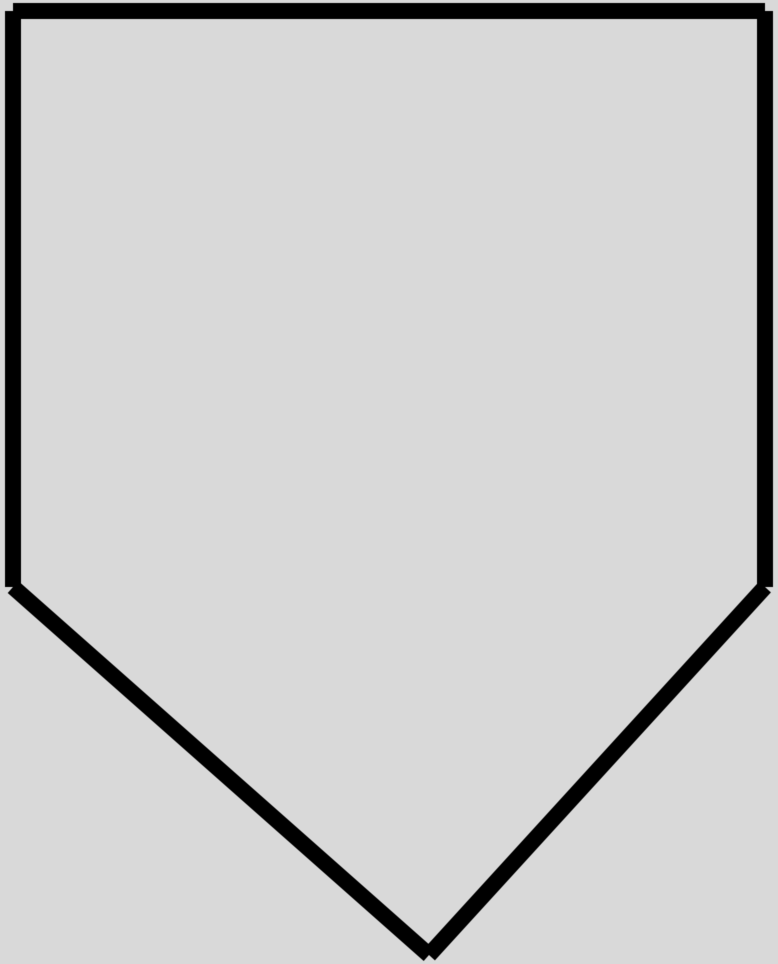
L5 diagnosis?



- Negative spring test
- B/L deep sulci
- B/L posterior ILAs

Work slide #10

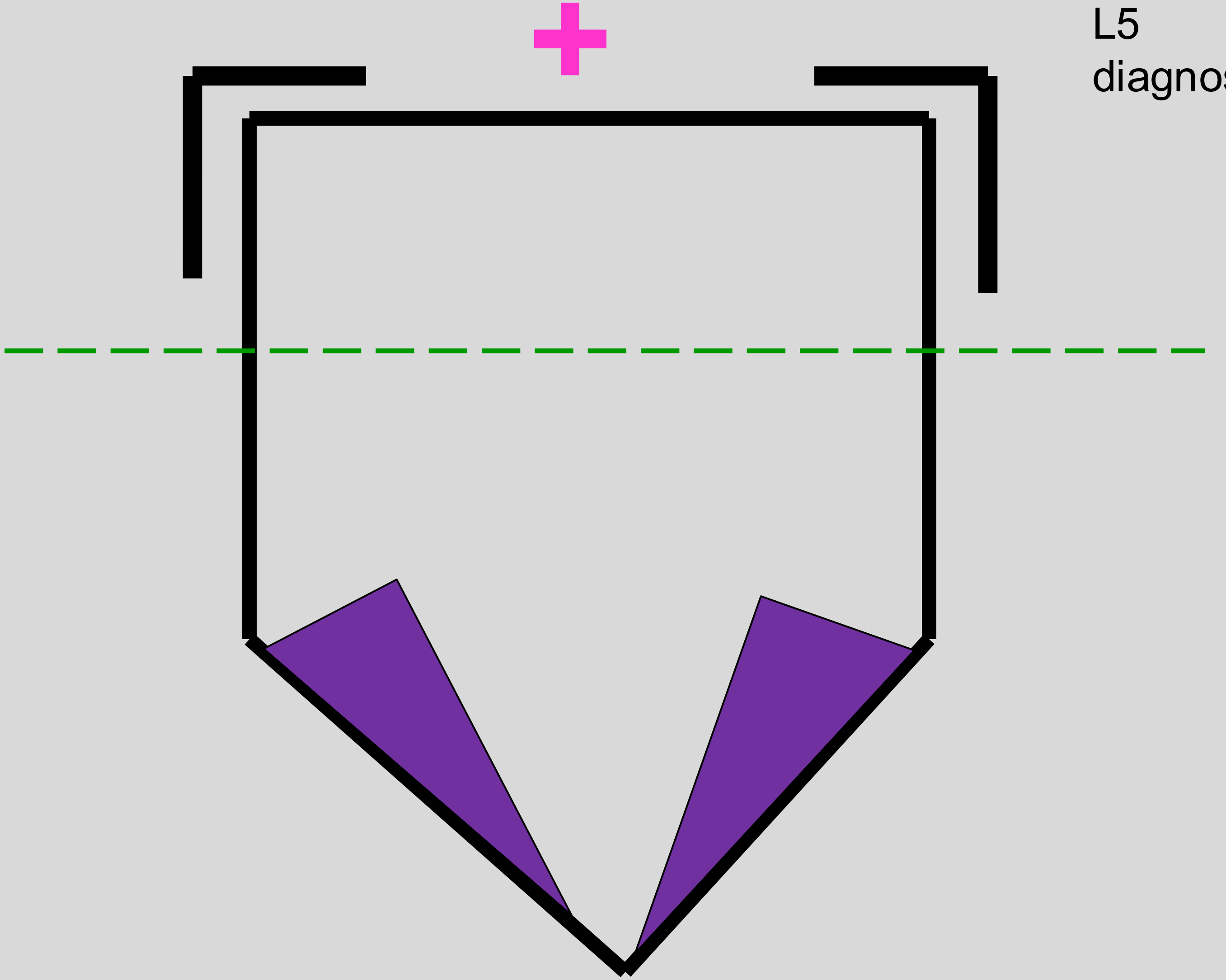
L5
diagnosis?



- positive spring test
- B/L shallow sulci
- B/L anterior ILAs

Work Slide # 10: **Bilateral Sacral Extension**

L5 diagnosis?



- positive spring test
- B/L shallow sulci
- B/L anterior ILAs

Self Assessment Question #1

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A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 is rotated left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #1

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A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 is rotated left. What is your sacral diagnosis?

1. **R on R Forward Sacral Torsion**
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #2

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A 35-year-old butcher falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a positive sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 FRS left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #2

A 35-year-old butcher falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a positive sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 FRS left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. **R on L Backward Sacral Torsion**
5. Right Unilateral Sacral Flexion

Self Assessment Question #3

A 24-year-old man had a heavy box fall out of his closet and twisted and caught it awkwardly. He presents to your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a shallow sacral sulcus on the right, an anterior-superior ILA on the right and L5 rotated right. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. Left unilateral Sacral Flexion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Extension

Self Assessment Question #3

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A 24-year-old man had a heavy box fall out of his closet and twisted and caught it awkwardly. He presents to your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a shallow sacral sulcus on the right, an anterior-superior ILA on the right and L5 rotated right. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. Left unilateral Sacral Flexion
4. R on L Backward Sacral Torsion
5. **Right Unilateral Sacral Extension**

Self Assessment Question #4

A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a positive standing flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. ASIS elevated on left, PSIS lower an the left, pubic tubercles are about even. L5 is rotated left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #4

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A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a positive standing flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. ASIS elevated on left, PSIS lower an the left, pubic tubercles are about even. L5 is rotated left. What is your sacral diagnosis?

1. **R on R Forward Sacral Torsion (also a left posteriorly rotated innominate)**
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #6

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A 35-year-old song writer presents your office c/o low-back pain. On exam you find L5 extended, rotated and side-bent right. From this finding you make a likely sacral diagnosis. Which of the following would you expect to go along with your sacral diagnosis?

1. Negative Spring test
2. Posterior-inferior ILA on right
3. Left positive seated flexion test
4. Left Deep Sulcus
5. Left oblique axis

A 35-year-old song writer presents to your office c/o low-back pain. On exam you find L5 extended, rotated and side-bent right. From this finding you make a likely sacral diagnosis. Which of the following would you expect to go along with your sacral diagnosis?

1. Negative Spring test
2. Posterior-inferior ILA on right
3. **Left positive seated flexion test**

   (The dx is Left on Right Backward Sacral Torsion)
4. Left Deep Sulcus
5. Left oblique axis

Physcal Findings and their Sacral Diagnoses

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STATIC FINDINGS				TESTS				
ILA Posterior	Deep Sulcus	Shallow Sulcus	L5 Rotation	Seated Flexion	Lumbar Spring	Sphinx Test	DIAGNOSIS	AXIS
Right	Right	Left		Right	Negative	Negative	Right Unilateral Shear	
Left	Left	Right		Left	Negative	Negative	Left Unilateral Shear	
Right	Left	Right	Right	Left	Negative	Negative	Right on Right (R on R) Rotation	Right Oblique
Right	Left	Right	Left	Left	Negative	Negative	Right on Right (R on R) Torsion	Right Oblique
Left	Right	Left	Left	Right	Negative	Negative	Left on Left (L on L) Rotation	Left Oblique
Left	Right	Left	Right	Right	Negative	Negative	Left on Left (L on L) Torsion	Left Oblique
Right	Left	Right	Right	Right	Positive	Positive	Right on Left (R on L) Rotation	Left Oblique
Right	Left	Right	Left	Right	Positive	Positive	Right on Left (R on L) Torsion	Left Oblique
Left	Right	Left	Left	Left	Positive	Positive	Left on Right (L on R) Rotation	Right Oblique
Left	Right	Left	Right	Left	Positive	Positive	Left on Right (L on R) Torsion	Right Oblique
Even	Bilateral			Equivocal	Negative	Negative	Bilateral Sacral Flexion	Transverse
Even		Bilateral		Equivocal	Equivocal		Bilateral Sacral Extension	Transverse

DiGiovanna, EL & Schiowitz, S, An Osteopathic Approach to Dx & Tx, Lippincott W &W
Table 60-1

Algorithm of Sacral Diagnosis:

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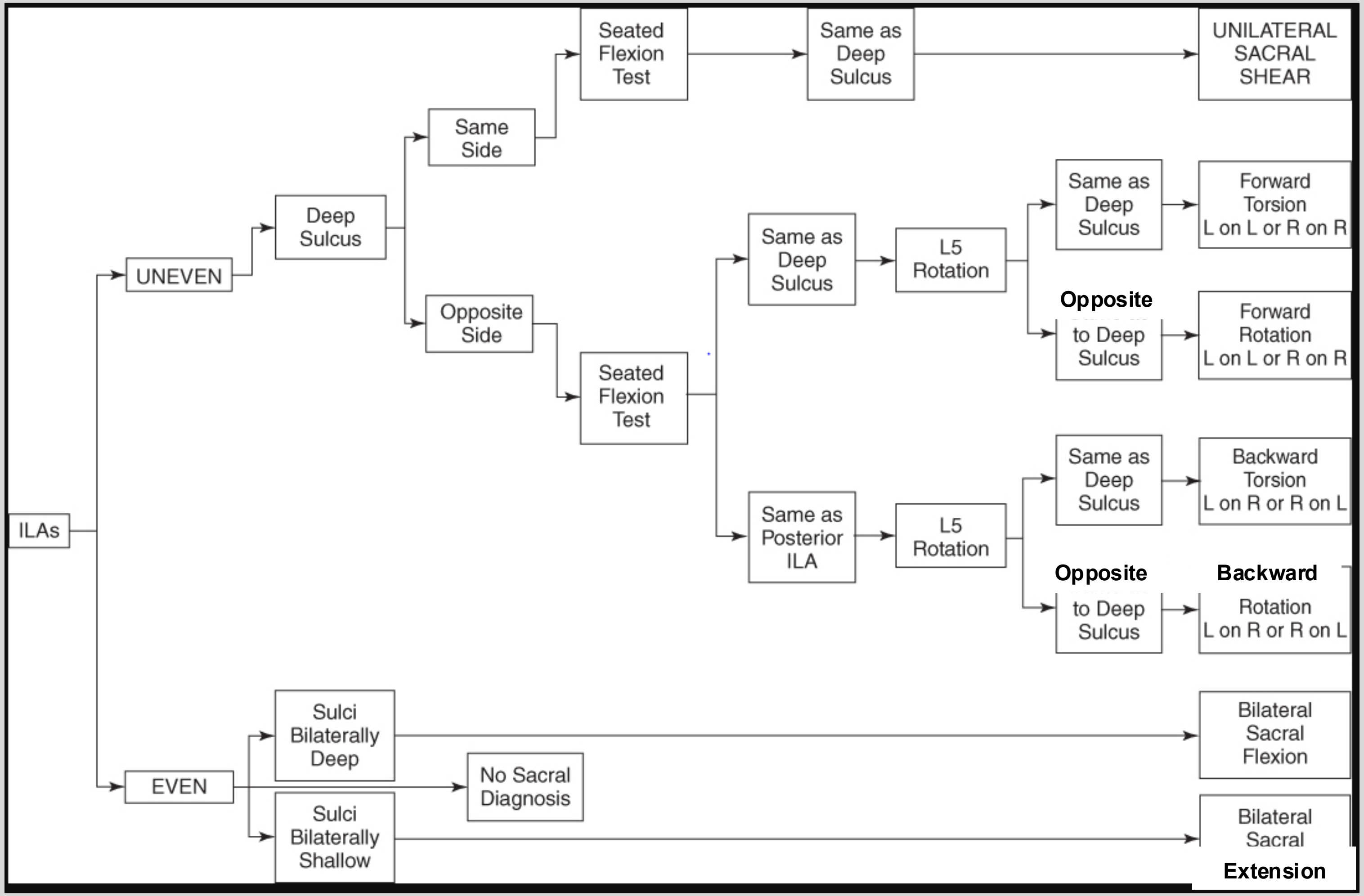

Table modified from: DiGiovanna 4e – Table 60-13

Summary Slide

- Static physical exam findings of the sacral region will be used to make the diagnosis of sacral somatic dysfunctions
- Diagraming your physical findings will help you in making the somatic dysfunction diagnosis

Session Objectives

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1. Be able describe the basic sacral landmarks and their locations
2. Be able to describe how to perform and interpret the different motion tests for sacral diagnosis.
3. Be able to describe the different sacral axes and their significances.
4. Be able to describe physiologic and non-physiologic sacral dysfunctions and what types of sacral dysfunction fall into each category and how they are named
5. Be able to apply the rules of L5 on the sacrum to make a sacral diagnosis if only given an L5 diagnosis.

Lumbosacral Junction

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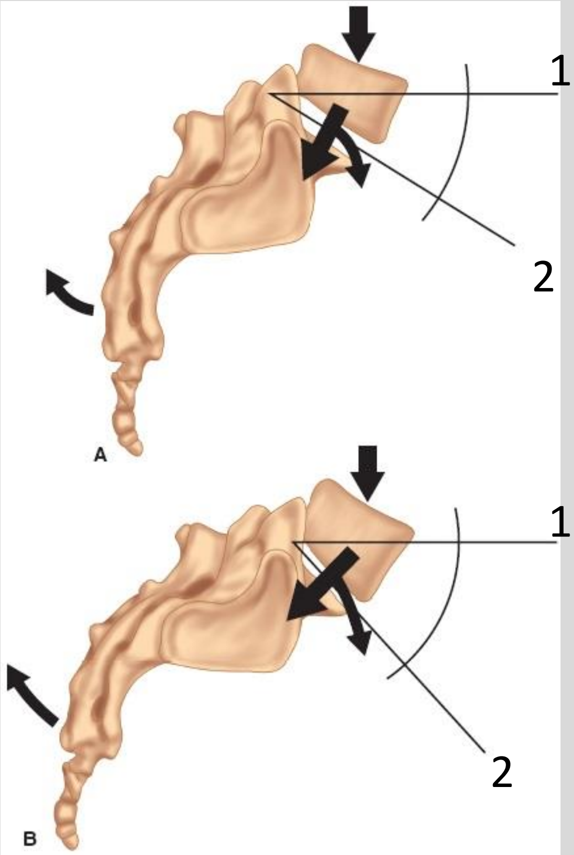

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Lumbosacral Angle (Ferguson's Angle) between:

1-horizontal

2-sacral base

Normal= Approximately 25°-35°

An increase in the lumbosacral angle creates a greater anterior vector force, which increases the lumbosacral strain.

Sacraliliac joint

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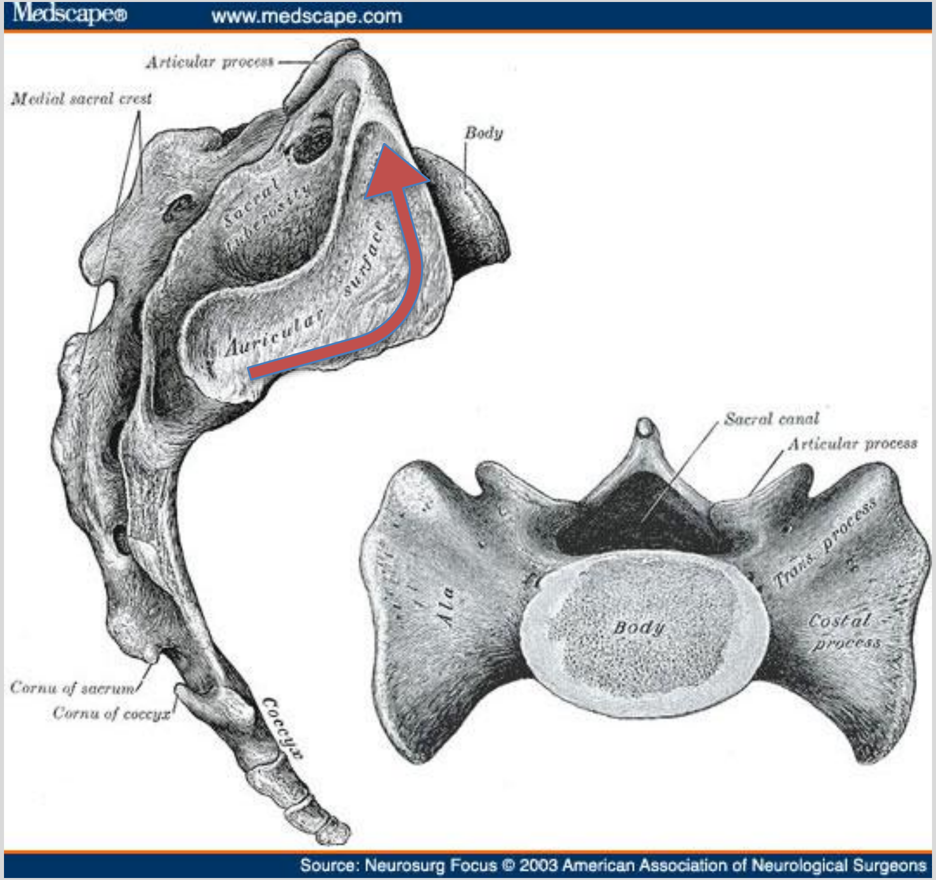
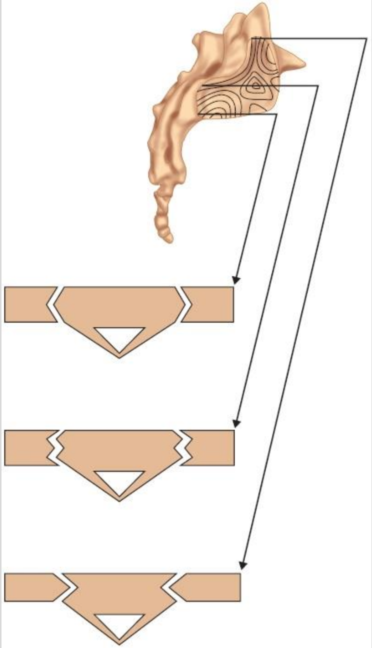


Sacroilac articulation configurations as seen at different sacral levels



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Somatic Dysfunction of the Sacrum may have an influence on Nerve structure and function

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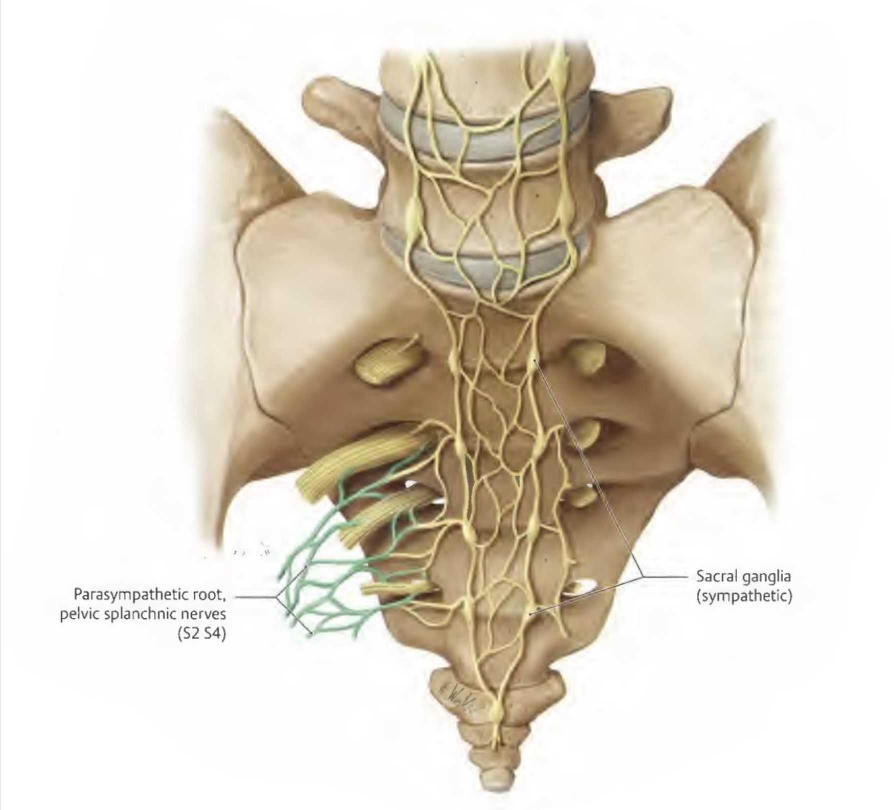
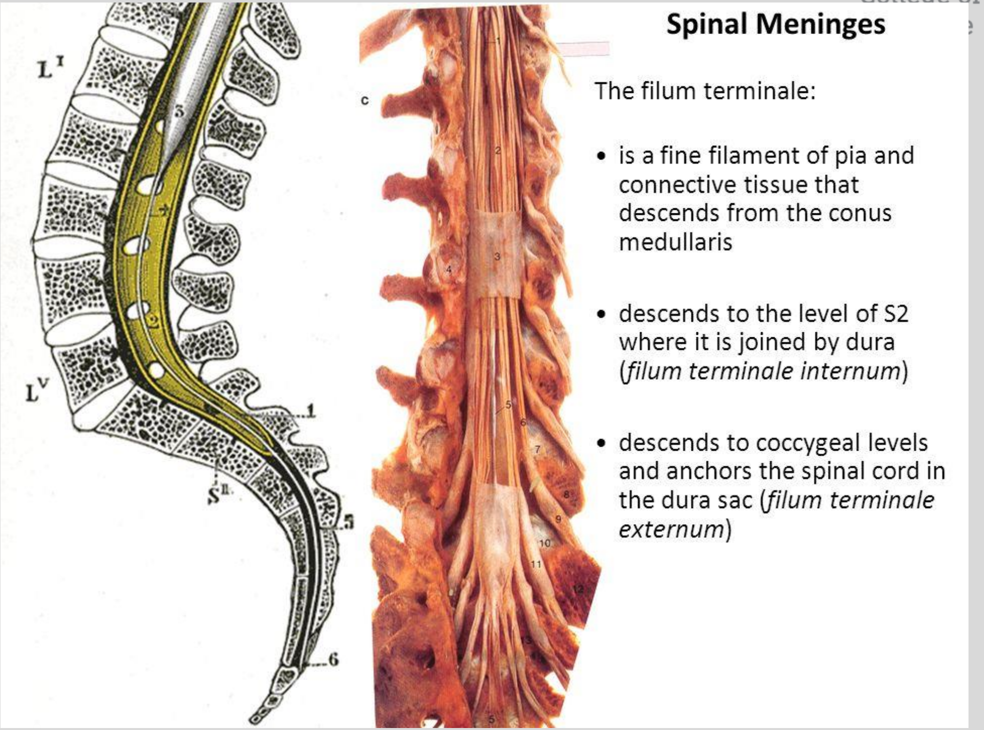
- Ventral nerve roots
- Parasympathetics - S2-4
- Sympathetic Chain Ganglia
  - Ganglion Impar @ sacrococcygeal junction

Image source unk



Caudal Dural Attachments

- S2
- Coccyx1



Piriformis muscle

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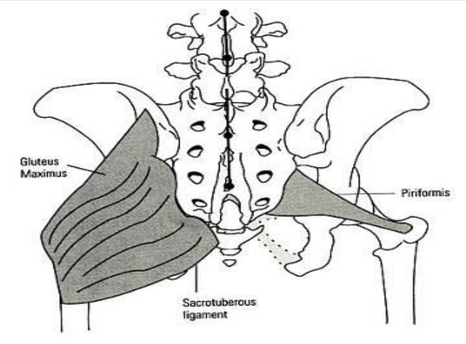
Origin/Insertion: Anterior sacrum to Greater Trochanter of Femur



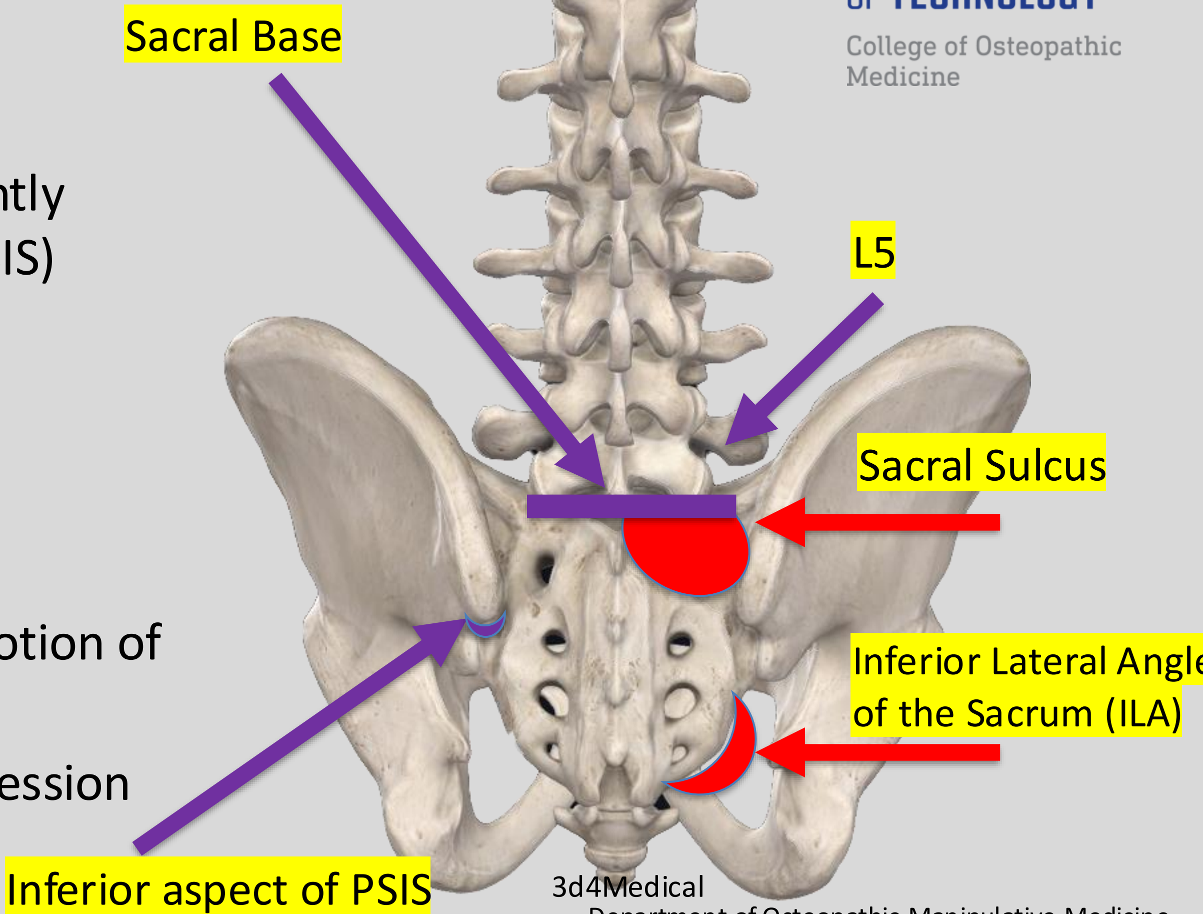
Action:

- External Rotation
- Extension of thigh
- Abduction of thigh if hip flexed

Nerve supply: S1,2

*In 10-11% of population the entire or peroneal portion of the sciatic nerve runs through the belly of piriformis, so if its hypertonic this may result in **"piriformis syndrome"**, pain radiating to posterior thigh, usually not below the knee

- These are the most important landmarks for making a sacral Diagnosis
  - Symmetry of:
    - Sacral Sulci (located slightly medil and superior to PSIS)
    - ILAs
  - Diagnosis of L5
  - Motion at:
    - Lumbosacral junction
    - SI joint with cephalad motion of ILAs
    - SI Joint with ASIS Compression and Seated Flexion test



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Axes of Motion of the Sacrum

1 - vertical axis

2 - right oblique axis (gait, sacral torsions and sacral rotation)

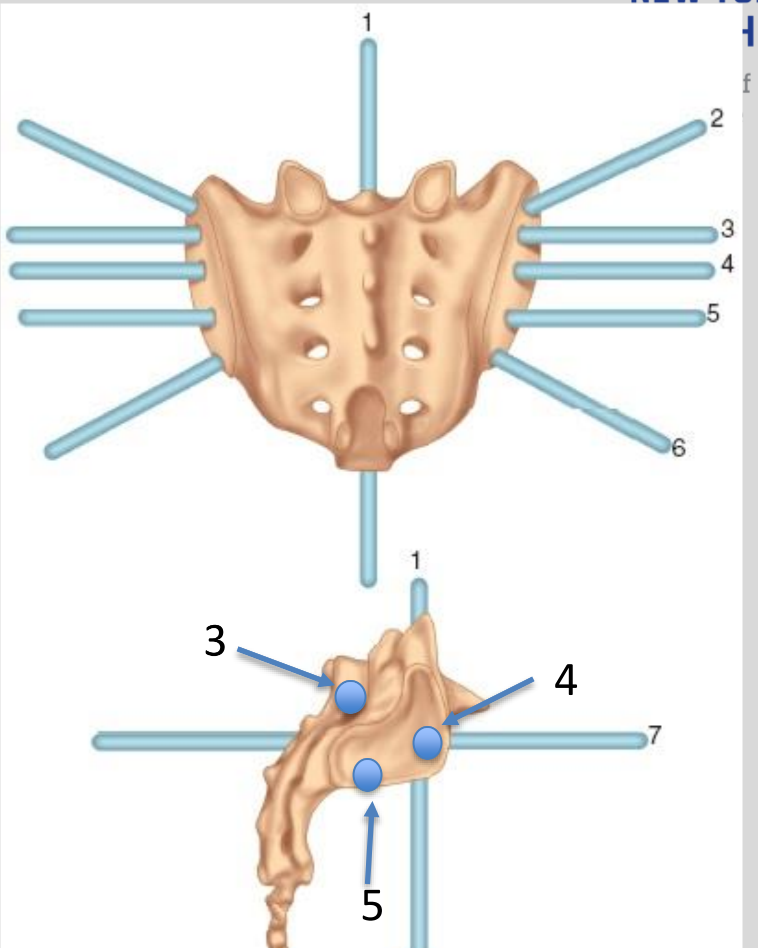
3 - Superior Transverse axis (respiratory and cranial-sacral motion)

*4 – Middle Transverse/*sacroiliac axis (Postural Motion)

5 - Inferior Transverse /iliosacral axis (Gait, Innominate rotation)

6 - left oblique axis (sacral torsions and sacral rotation)

7 - anteroposterior axis



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Nutation (nodding) vs. Counternutation

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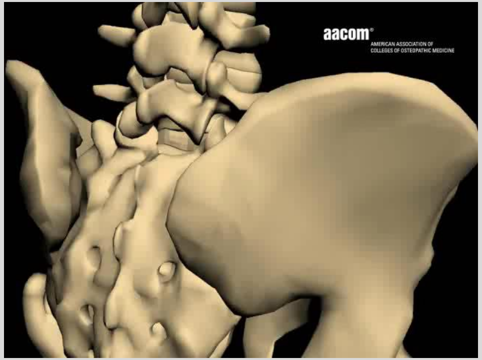
Axis: Middle Transverse

Other ways to say these:

- Anterior Nutation vs. Posterior Nutation
- (Sacral) Base Anterior vs. Base posterior
- Bilateral sacral flexion vs. B/L sacral extension



Sacral Biomechanics

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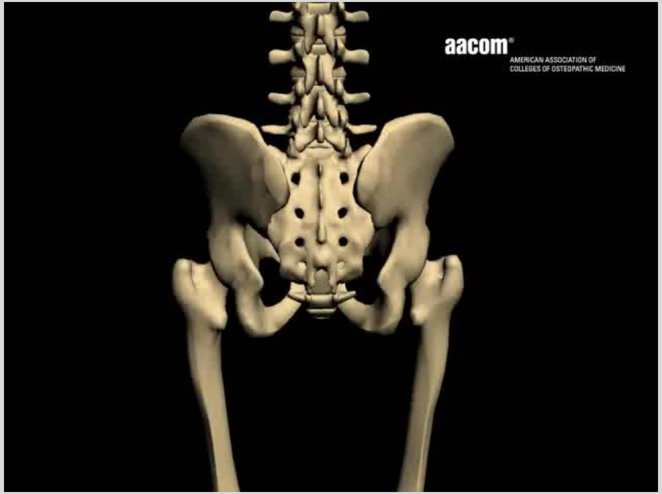
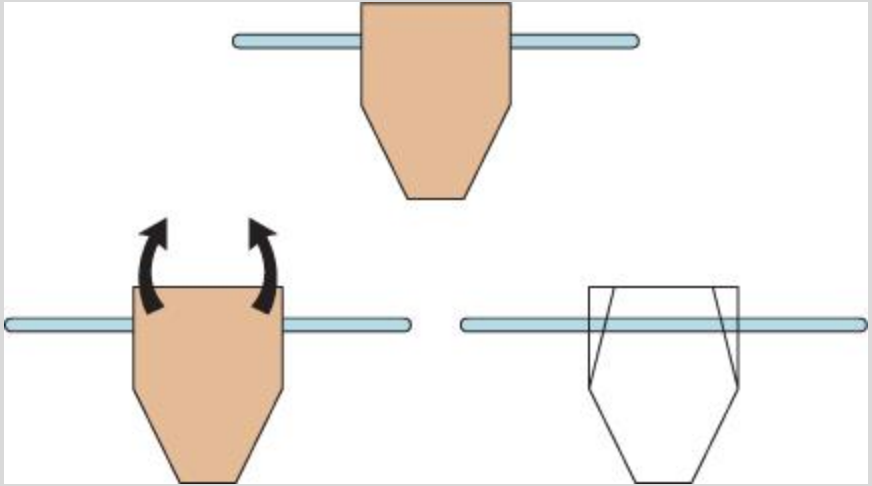
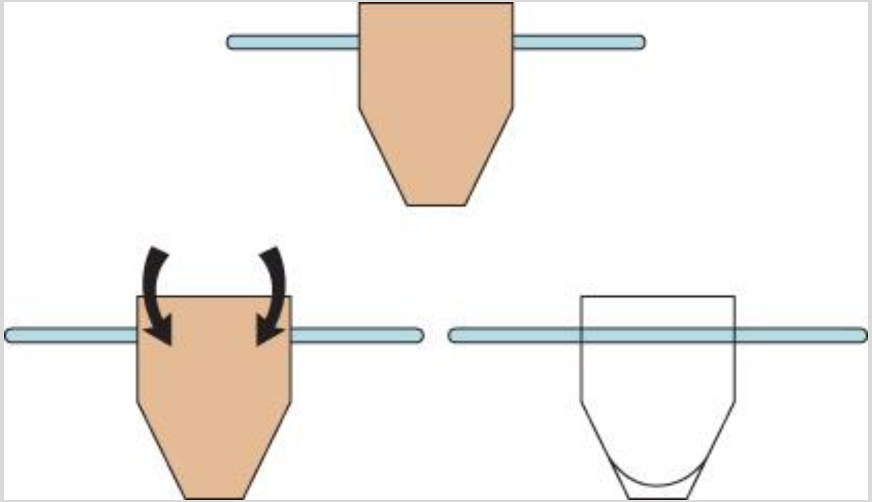


Bilateral Movement around a Transverse Axis



Sacral base is bilaterally flexing over a transverse axis



Sacral base is bilaterally extending over a transverse axis

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Unilateral Movement seems to move around a "Transverse Axis?"

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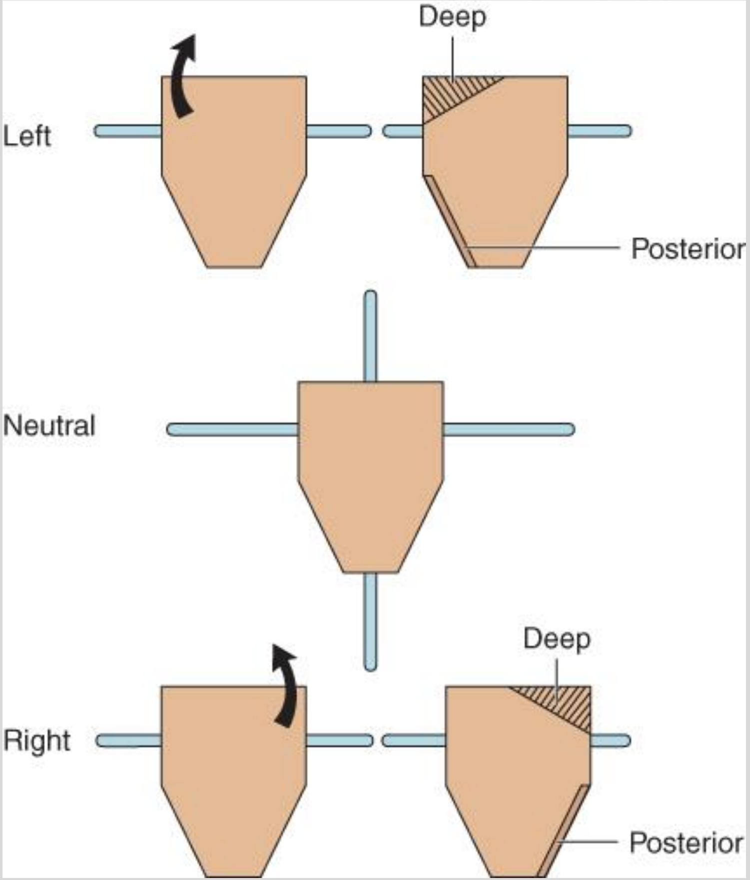
The sacrum can flex or extend unilaterally. The axis for this motion are not well understood. It is <u>not</u> really around a transverse axis. From a palalpatory standpoint, the following occurs. In unilateral flexion, the sacral base on the side of flexion will move anterior and the ILA on the same side will move posterior (and a little inferior). The opposite occurs in unilateral extension



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Oblique Sacral Axes

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Left Oblique Axis

- Starts from the left superior sacroiliac articulation
- Diagonal to right inferior SI articulation

Right Oblique Axis

- Starts from the right sacral sulcus
- Diagonal to left infero-lateral angle of sacrum

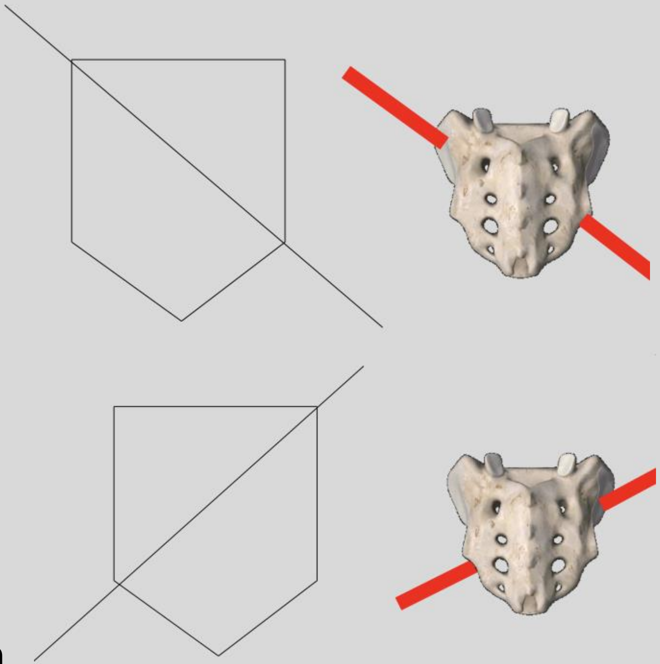
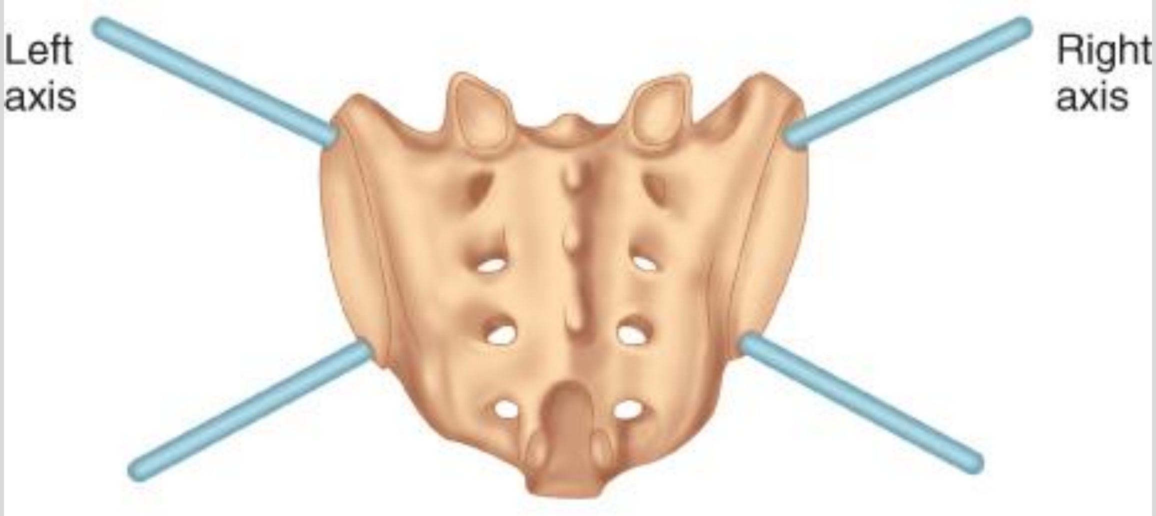


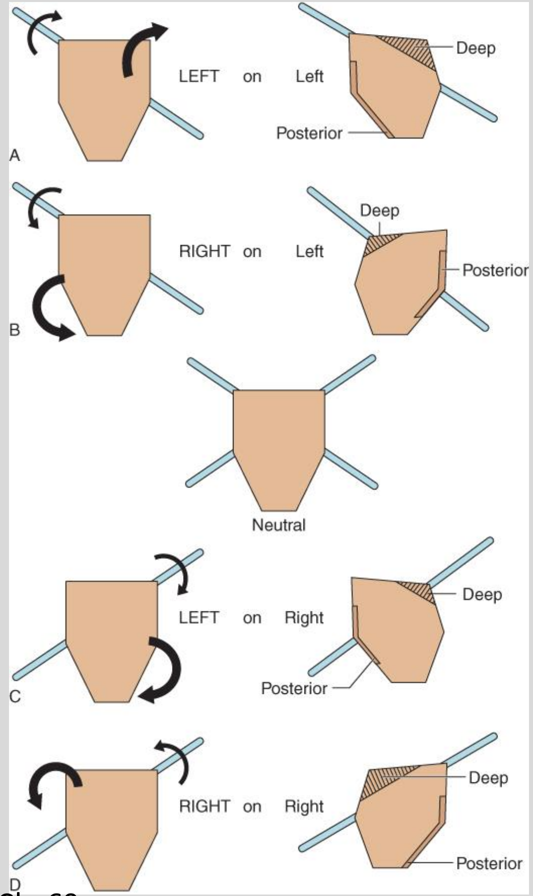
Image: 3D4Medical.com

Neutral

Movement around an oblique axis

Examples are left rotation around a left oblique axis and left rotation around a right oblique axis





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Example: Left Sacral Rotation over oblique axes

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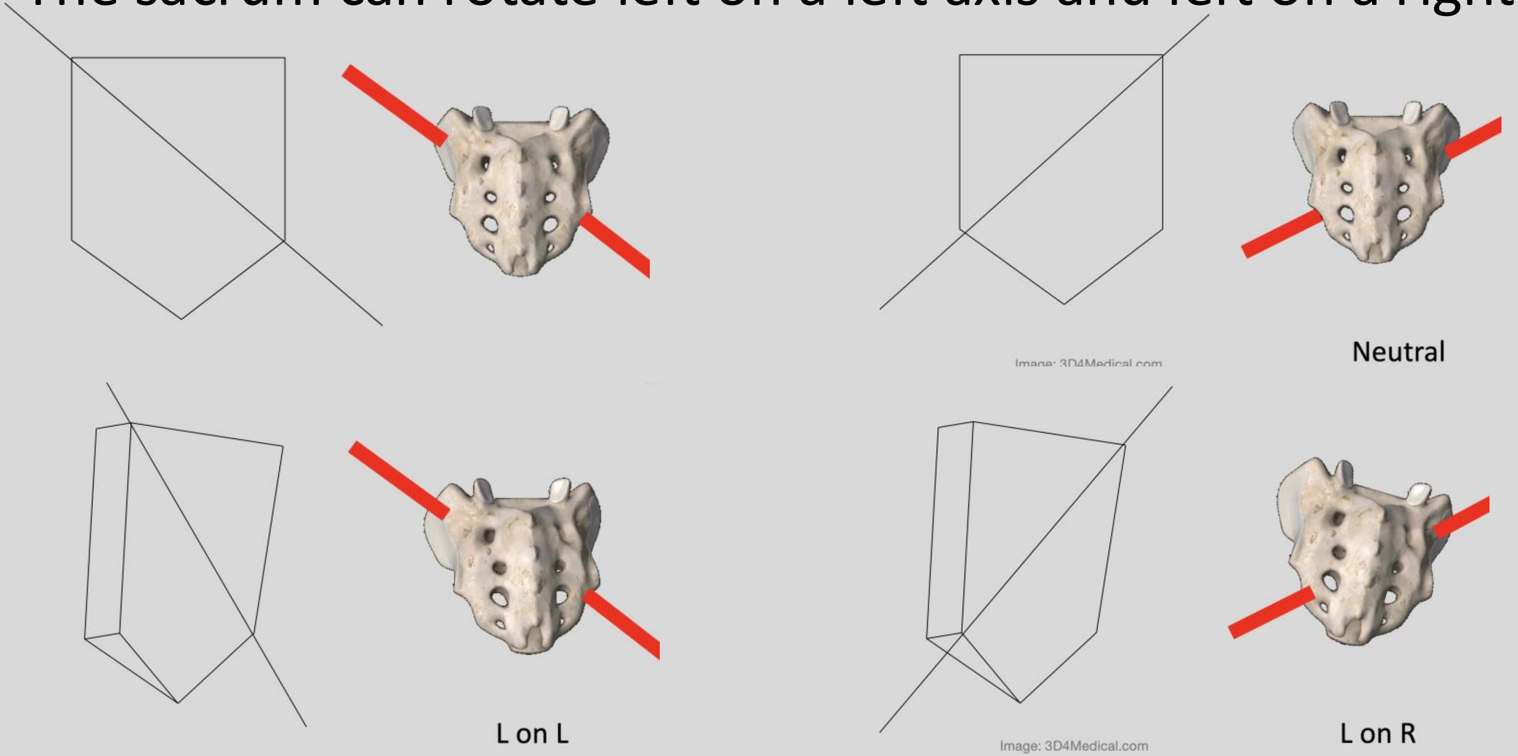
- The sacrum can rotate left on a left axis and left on a right axis



Osteopathic Diagnosis of the Sacrum

Prone Static Landmark Examination

+

Spring or Sphinx Test

+/-

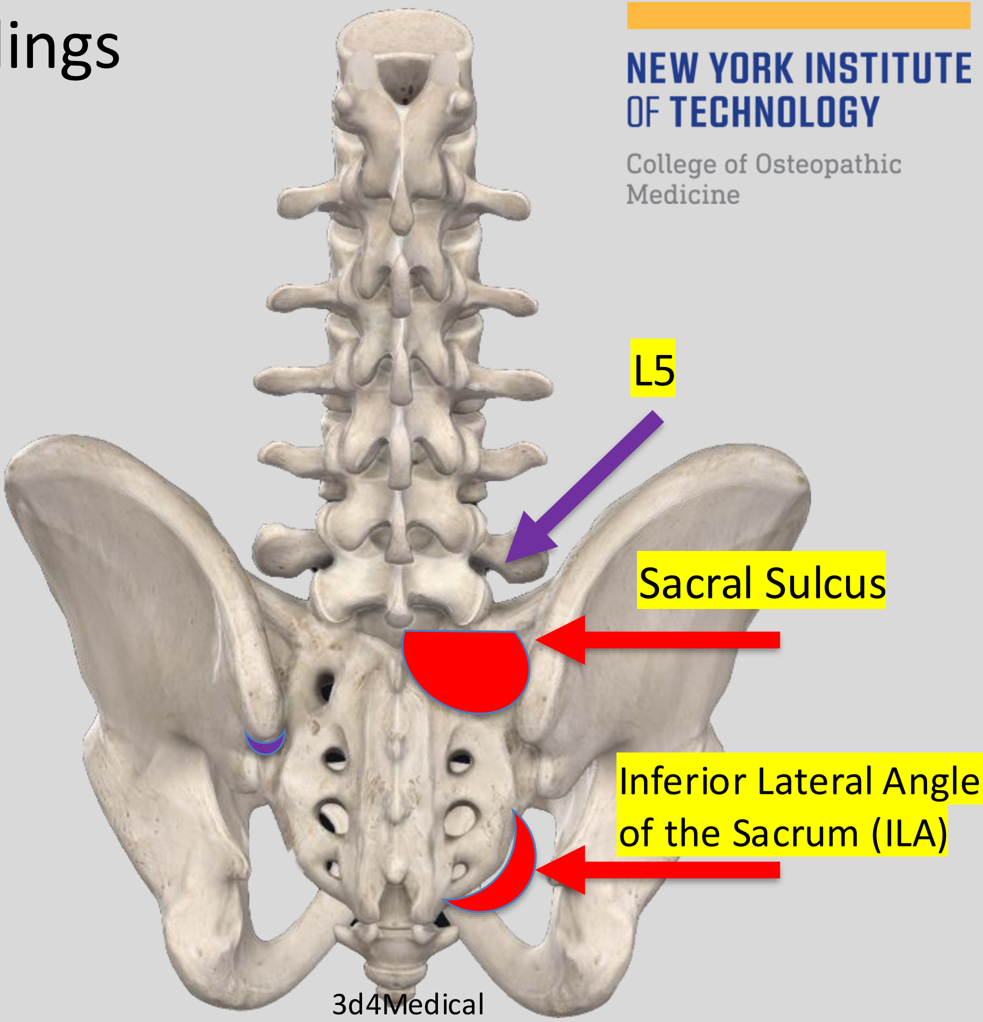
Seated Flexion Test

Physical Exam: Prone Static findings

1. L5

2. Sacral Sulci

3. Inferior Lateral Angles (ILAs)



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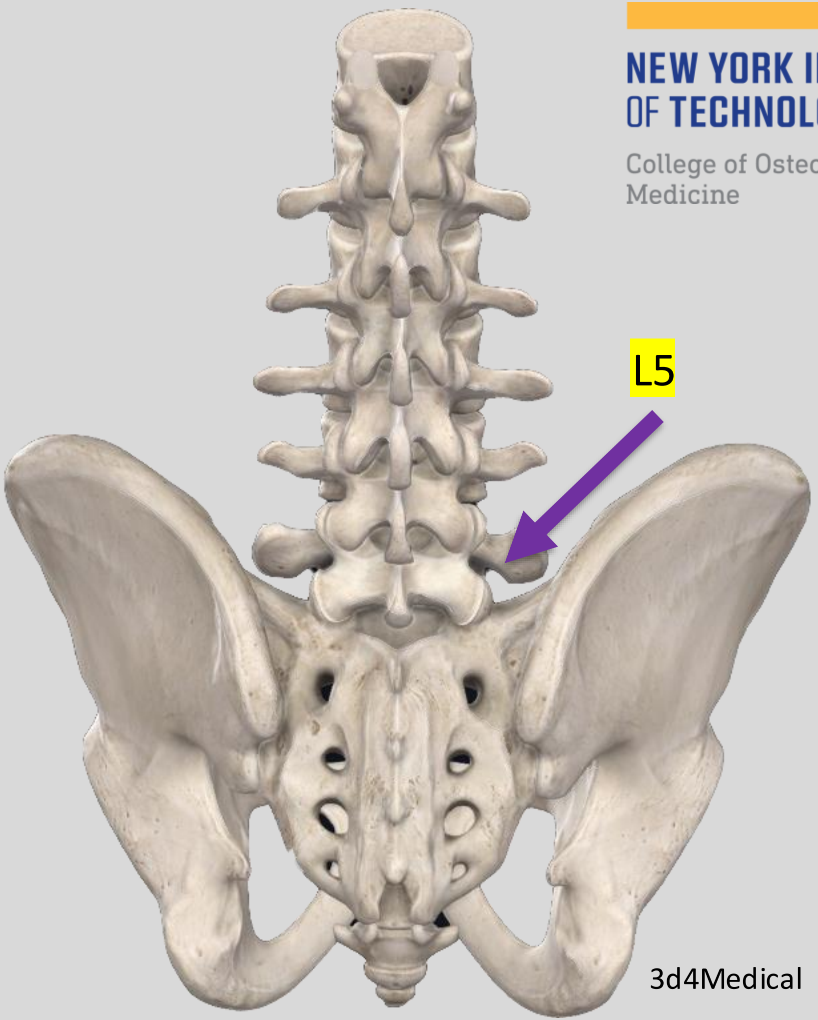
- Diagnose L5
  - Direction of L5 rotation influences certain sacral diagnoses:
  - Sacral Torsion Somatic Dysfunctions:
    - ❖ L5 & Sacrum rotate opposite
  - Sacral Rotation Somatic Dysfunctions:
    - ❖ L5 & Sacrum rotate same direction



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Check the symmetry of the Sacral Sulci

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- Note which side is deeper (sacral base on that side is more anterior) & note which side is more shallow (sacral base on that side is more posterior)

- *Patient position:* prone.
- *Physician position:* varies according to the region examined.
- *Sacral sulci*
    - The physician stands at the side of the table and faces toward the patient's head.
    - The physician places her thumbs on each posterior superior iliac spine (PSIS).
    - The physician hooks her thumbs medially along the PSIS and brings her finger pads into the sacral sulci bilaterally
- The physician evaluates the relative depth of the sulci by two means: By palpating the depth of each sulcus by thumb position.
- By lowering her eyes to the level of the sulci for visual evaluation of depth.

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Check symmetry of ILAs

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- Note which side is posterior and inferior.
    - Remember if one side is posterior and inferior, the other side is anterior and superior.

1. Patient Prone
2. Physician's thumbs on inferior aspect of ILAs.
3. Compare which side is inferior
4. Then pace thumbs on posterior aspect of ILAs
5. Compare which side is more posterior

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Lumbosacral Spring Test

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- **Must be on L-S (lumbo-sacral) JUNCTION**
- **Be mindful of patient**
- **Gentle anterior force**

NORMAL SPRING TEST: If there is spring-ability at the L-S junction this is physiologic= a "*negative*" test:

- Forward moving sacral base (such as seen with forward torsions/rotations, and B/L sacral flexions)

ABNORMAL SPRING TEST = A positive (+) Test = ABSENCE of spring-ability :"steel-like resistance" – non-physiologic. Often result of sacral base being extended or backward

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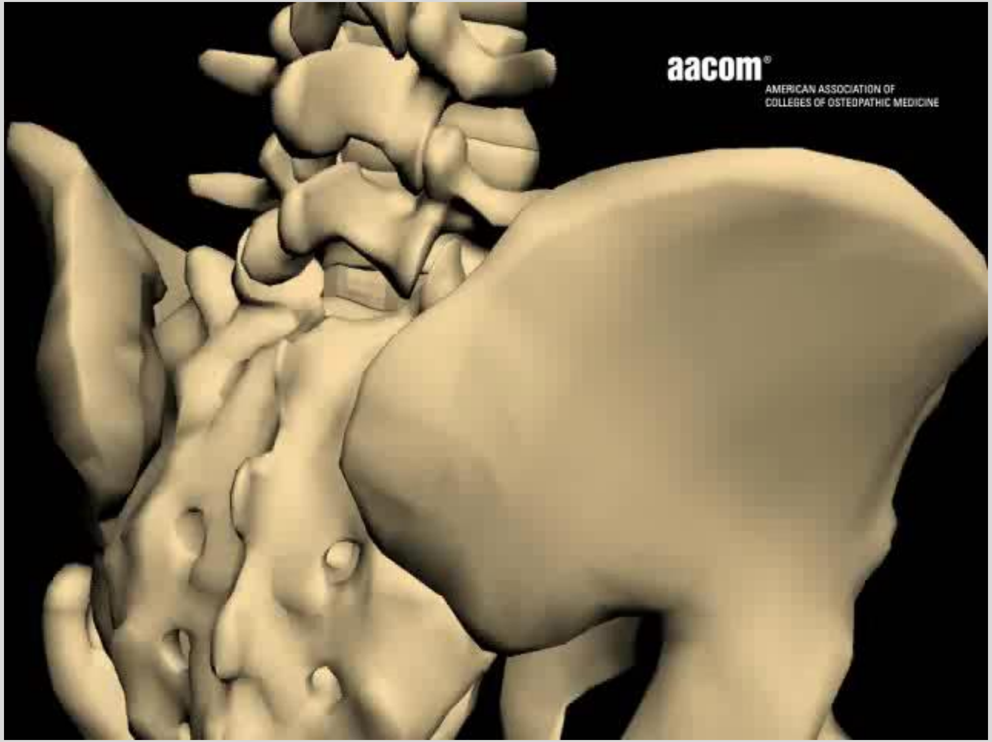
Sacral Biomechanics



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Sphinx test (backward bending test)

1. Interchangeable with Spring Test

Step 1: Patient prone – make note of static findings (sulci and ILAs)

Step 2: Physician places fingers at sulci and thumbs at ILA's B/L

Step 3: The patient is then instructed to either raise her body onto her elbows or if necessary onto palms to induce extension at the L-S junction. There should be an increased lumbar lordosis and sacral base should go anterior B/L during back extension.

(+) Test = Asymmetrical positioning of ILAs is maintained or worsened. This indicates a NON-physiologic sacral Somatic Dysfunction

(-) Test = ILA's become more symmetric when the patient's back is in extension.

Sphinx test (backward bending test)

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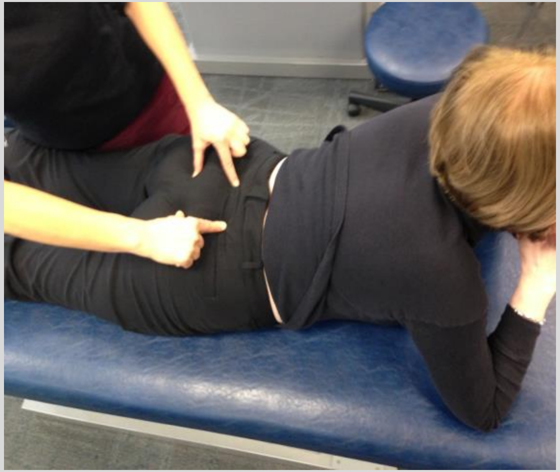

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Seated Flexion Test

1. Patient sits with **feet flat on the floor**, knees parted
2. Physician is eye level with patient's PSIS's
3. Physician's thumbs are placed gently on inferior aspect of PSIS bilaterally
4. Patient bends forward from the waist with their arms dropping between their legs (not resting on their thighs)

(+) Test = On the side that your thumb moves FIRST & FURTHEST
Indicates Sacral ON Innominate somatic dysfunction (SD)

- **Side of positive test is used only in naming of (ipsilateral) unilateral sacral category of SD**
- **Side opposite of positive test is used for naming the axis for sacral torsions and rotations**
- But, the patients don't always follow the rules…will explain in lecture

(-) Test = Either: a) No SD

OR

b) Bilat. SD (B/L flexion or B/L extension SD)

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ASIS compression test

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1. Patient is supine
2. Physician places her thenar/ hypothenar eminences along ASIS's B/L
3. Physician alternately introduces gentle but firm downward (A-P) pressure toward one S-I joint at a time from its ipsilateral ASIS contact. Physician allows the ilia to recoil anteriorly against gentle pressure
4. Determine ease of motion and compare bilaterally

(+) Test = Side that offers greater resistance to compression (left or right)

*ASIS Compression test is **NOT** specific for ilio-sacral vs. sacro-iliac dysfunction.

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ILA Motion Testing

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(Another Test of Sacroiliac Joint (SIJ) Somatic Dysfunction Laterality)

- Place thenar eminences on the inferior part of ILA's and fingertips at sulci
- Direct force cephalad from ILA into ipsilateral SIJ
- (+) test indicating dysfunction: a decrease in joint play on one or both sides and restriction of sacral motion.
- NOT used for naming sacral dysfunctions but a method of seeing which side may be restricted

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Physiologic vs. non-physiologic Categories of Sacral Sdysf(x)

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- **<u>Physiologic Dysfunction</u>**- somatic dysfunction produces a restriction

within the normal range of motion of a joint

Examples:

  - Forward Sacral torsions
  - Forward Sacral rotations
  - Bilateral Sacral Flexions

- **<u>Non-physiologic dysfunction</u>**-restriction of a joint in a position not obtained by usual physiologic motion. Usually traumatic but may occur in a patient with hypermobility

Examples:

  - Backward sacral torsions
  - Backward sacral rotations
  - Unilateral Shears (sacral flexions/extensions)
  - Bilateral Sacral Extensions

Non-physiologic dysfunctions should be treated prior to attempting OMT to other diagnoses

Physical exam → diagnosis

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1. Seated flexion test (Right or Left or neither?)
2. Sphinx test or Spring test (positive or negative?)
3. L5 Diagnosis (Type 1 or 2? SB? Rotation?)
4. Deep Sacral Sulci (R, L, both, neither?)
5. Posterior/Inferior ILA (R, L, both, neither?)

You will learn how to putthese findings into specific Diagnoses in the lab lecture and lab session.

DIAGNOSING SACRAL <u>TORSIONS</u> BASED ON L5 (You will need to memorized this slide and the next 3 slides. This is a different way of diagnosing sacral torsions which you may see on future exams. In the Lab, we will be going over how to diagnose sacral dysfunctions using the sacral landmarks and and other findings as discussed earlier in this presentation.)

- **L5** will be **rotated opposite** the direction of the **sacrum (if not, L5 must be treated first)**
- **Seated flexion test is (+) opposite the axis of rotation** in a torsion**.**
- **L5** will be **sidebending** to the same side as the oblique **Axis**

Session Objectives

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1. Be able describe the basic sacral landmarks and their locations
2. Be able to describe how to perform and interpret the different motion tests for sacral diagnosis.
3. Be able to describe the different sacral axes and their significances.
4. Be able to describe physiologic and non-physiologic sacral dysfunctions and what types of sacral dysfunction fall into each category and how they are named
5. Be able to apply the rules of L5 on the sacrum to make a sacral diagnosis if only given an L5 diagnosis.